Tunnelling and Underground Space Technology 168 (2026) 107207

Contents lists available at ScienceDirect

Tunnelling and Underground Space Technology incorporating Trenchless Technology Research

journal homepage: www.elsevier.com/locate/tust

Intelligent technologies for tunnel construction and maintenance: A state-of-the-art review of methods and supporting platforms

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ARTICLE INFO

Keywords:
Urban underground space
Shield tunnel
Construction and maintenance
Intelligent platform
Carbon neutralization

ABSTRACT

With the rapid integration of the Internet of Things (IoT), Big Data, and Artificial Intelligence (AI), tunnel engineering is undergoing an accelerated shift toward intelligent construction and operation & maintenance (O&M). This paper systematically reviews the technological evolution of intelligent tunnel systems across design, construction, and operation & maintenance (O&M) phases. In the design stage, emphasis is placed on digital design tools and geological digital twins to enhance constructability. Intelligent perception, real-time risk analysis, and autonomous control jointly improve construction safety and efficiency. In the O&M phase, real-time adaptive monitoring combined with predictive maintenance ensures sustained tunnel performance and resilience. Parallelly, the development of tunnel information platforms has transitioned from standalone systems to integrated, modular, and interoperable platforms that enable multi-source data fusion, collaborative management, and real-time decision-making. This review highlights key enabling technologies, practical applications, and persistent bottlenecks, including data heterogeneity, cost-efficiency, and model generalization. Looking forward, intelligent platforms are expected to evolve into closed-loop ecosystems that combine real-time sensing, AI-based optimization, and digital twin co-simulation. These systems will also embed sustainability indicators such as carbon tracking, material recycling, and lifecycle assessment to support green decision-making. Collectively, the study provides a comprehensive roadmap for developing next-generation intelligent, resilient, and sustainable tunnel systems.

1. Introduction

With the rapid advancement of Internet technologies, the Internet of Things (IoT), and digital transformation, urban rail transit infrastructure is undergoing a new wave of intelligent upgrading. In the face of increasingly complex tunnel construction environments and rising demands for efficient, safe, and sustainable engineering, the intelligentization of tunnel construction and O&M processes has become an inevitable trend. Rather than implying that tunnels themselves are 'intelligent', this trend refers to the integration of intelligent sensing, big data analytics, and automated control technologies into LCA, from design and construction to O&M. Chen et al. (Chen et al., 2022) and Zhu et al. (Zhu et al., 2024) have jointly pointed out that the development of intelligent tunnel construction lies in the deep integration of information technology with engineering systems, enabling full-process digital empowerment, intelligent sensing, and automated control to foster resilient and high-quality tunneling.

Recent studies have increasingly attempted to define the scope and technical foundation of 'intelligent tunnel construction.' Qin (Qin et al., 2024b) systematically summarized the key technological components, ranging from rock-machine perception to digital platform integration and proposed a framework that supports the evolution from traditional automation to full intelligentization, bridging tunnel construction and O&M phases. However, the O&M stage continues to face critical challenges, including heterogeneous data integration, real-time structural monitoring, equipment degradation, and the absence of robust decision-

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https://doi.org/10.1016/j.tust.2025.107207
Received 29 May 2025; Received in revised form 26 September 2025; Accepted 21 October 2025
Available online 29 October 2025
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support mechanisms for maintenance optimization. To mitigate these issues, emerging intelligent monitoring platforms have begun incorporating IoT, sensor networks, and big data analytics to enable condition-based maintenance and proactive risk mitigation.

Scholars have explored various approaches to advancing intelligent tunnel construction and O&M. Padgett (Padgett et al., 2024) highlighted the need for dynamic, adaptive systems that address uncertainty while incorporating social equity considerations. Bai (Bai et al., 2020) designed a three-tier architecture for intelligent O&M platforms in urban rail systems, integrating big data, BIM, and machine learning to enhance lifecycle management. In parallel, several review articles have analyzed intelligent technologies from specialized perspectives (Chen et al., 2025; Liu et al., 2025a; Liu et al., 2024a; Xie et al., 2025a). For instance, Chen et al. (Chen et al., 2023b) provided a comprehensive review of seismic resilience assessment frameworks for shield tunnels, outlining current research progress, challenges, and future directions in tunnel resilience modeling and performance-based design. Zhao et al. (Zhao et al., 2024c) emphasized lifecycle-based digital twin frameworks for subway tunnels. Song et al. (Song et al., 2024b) examined digital early-warning platforms in mining, offering transferable insights for tunnel risk management. Liu et al. (Liu et al., 2024a) focused on AI integration and conducted a bibliometric analysis evaluating machine learning methods combined with multi-index evaluation frameworks, and Song et al. (Song et al., 2025) systematically examine the application progress of deep learning in tunnels and underground engineering across seven key areas. Rather than restricting attention to a single tunneling method, this review adopts a broad perspective that spans shield tunnels, drill-and-blast tunnels, mining tunnels, immersed tunnels, and urban utility tunnels serving transportation and infrastructure needs. Drawing on publications indexed in the Web of Science (WOS) over the past five years, this paper provides a comprehensive synthesis of recent progress.

This review systematically examines technological advances in intelligent tunnel engineering, focusing on sensing, data-driven modeling, automated control, real-time risk warning, and integrated platform development. By clarifying the role of intelligent technologies across the full tunnel lifecycle, covering design, construction, and O&M, it provides a holistic perspective that links technological evolution, platform integration, and sustainable development. Through this lifecycle-oriented framework, the review not only assesses current progress but also highlights key challenges and outlines future directions. The overall technical framework is illustrated in Fig. 1.

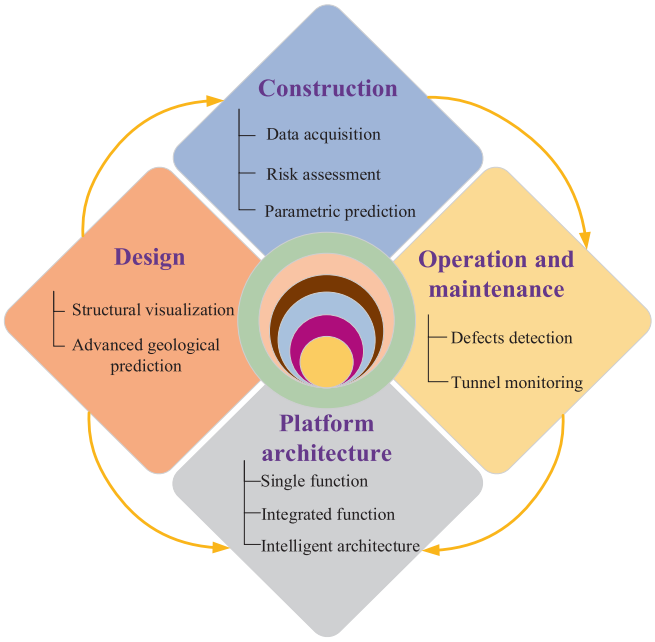

Fig. 1. Framework of this review.

2. Key technologies for intelligent tunnel construction and O&M

Tunnel structures are subjected to long-term influences from geological evolution, traffic loads, and environmental factors, which may lead to deterioration, cracks (Xu et al., 2025), settlement (Song et al., 2024a), and other structural problems. Scientific and technical inspections and assessments are essential throughout the tunnel lifecycle to ensure safety. During the design phase, tunnel intelligence platform offers accurate data support and optimization solutions through advanced geological prediction technologies and structural visualization tools. In the construction phase, it ensures safety and quality control (Ma et al., 2024) through real-time data acquisition, risk identification, and intelligent decision-making technologies (Zhang et al., 2025a). In the O&M phase, the platform acts as a unified center for integrating data, facilitating visualization, and supporting intelligent analytics. Drawing on digital twin frameworks and AI-driven analytics, intelligent platforms support long-term structural performance evaluation, surface defect detection, and predictive maintenance decision-making, and also predicts potential risks with high accuracy and provides targeted maintenance strategies (Zhang et al., 2024c; Zhou et al., 2024a). By applying these technologies, intelligent O&M through the lifecycle of tunnels can be achieved. This approach significantly enhances operational efficiency and safety, reduces the potential risks, extends tunnel utilization, and provides vital technical support for the construction of smart and resilient cities.

In this section, the classification of intelligent technologies into design, construction, and O&M phases is based on their primary functional objectives and typical temporal application within the tunnel lifecycle.

2.1. Application of key technologies in the tunnel design phase

Traditional tunnel design has largely depended on two-dimensional drawings and static documentation, which limits integration with dynamic, data-driven workflows and constrains efficiency, precision, and adaptability in modern tunnel engineering. To overcome these shortcomings, intelligent design and analysis systems have emerged, enabling the integration of geological surveys, structural performance requirements, and environmental constraints within advanced computational frameworks. Coupled with 3D visualization interfaces, these systems support performance-based and inverse design approaches and promote seamless transitions from digital models to construction planning and operation and maintenance. This section reviews the development of intelligent technologies in the tunnel design phase, with emphasis on structural visualization and advanced geological prediction. Representative studies were synthesized based on a systematic search of Web of Science using keywords such as ‘intelligent design’, ‘structural visualization, and ‘geological prediction’.

2.1.1. Structural visualization

Structural visualization during the tunnel design phase supports designers, engineers, and other stakeholders in intuitively understanding and effectively communicating the spatial configuration, functional characteristics, and potential challenges associated with tunnel structures. With the advancement of intelligent construction technologies, visualization methods have evolved beyond static 3D representations, becoming integrated platforms for multi-source data fusion, predictive analysis, and proactive risk management.

Building Information Modeling (BIM) has become a central tool in the digital transformation of tunnel infrastructure, offering a unified platform to represent both physical structures and functional systems. Widely adopted in the design phases of large-scale metro and highway tunnel projects in China and abroad, BIM enables the development of

detailed 4D models that integrate spatial configurations, structural elements, and utility networks, as shown in Figs. 2 - 3. It supports automated clash detection and cross-disciplinary coordination, thereby minimizing design conflicts and construction delays. Mature commercial software platforms (Revit, Civil 3D) have ensured the practical applicability of BIM in typical urban tunneling contexts. However, challenges remain in terms of data interoperability, especially when integrating BIM with GIS, IoT monitoring systems, or digital twin platforms during operation and maintenance stages. These limitations can hinder large-scale, real-time applications across multi-phase tunnel lifecycle management.

In recent developments, the role of BIM has further expanded to support intelligent design by integrating predictive modeling, real-time monitoring, and risk visualization functionalities. For example, in a case study of the Loushan tunnel, He (He et al., 2025) combined Variational Modal Decomposition (VMD) with Bidirectional LSTM (BiLSTM) to predict surrounding-rock deformation. The prediction outputs were embedded into a BIM plugin, with graphs and color-coded warnings enabling early identification of vault settlement risk. Another study proposes using System Theoretical Process Analysis (STPA) together with the VIKOR method to evaluate ground settlement risk in shield tunnelling, and suggests spatial visualization of risk levels within a BIM environment (Chen et al., 2024b). Despite these promising advances, their deployment remains largely limited to pilot or research projects. Key challenges include the reliance on high-quality monitoring data and difficulties in integrating heterogeneous sources into BIM systems. Moreover, the scalability of such predictive applications to large tunnel networks and real-time O&M scenarios remains limited.

In addition to AI-based predictive modeling and risk visualization, BIM platforms have also been increasingly integrated with physics-based numerical simulations to support dynamic decision-making during tunnel construction. These simulation approaches have been increasingly integrated into BIM platforms, enabling synchronized model updates and real-time data visualization (Cheng et al., 2025). Current practices have also incorporated 3D geological models, geotechnical parameters, and tunnel lining design details into BIM systems. Through advanced parametric modeling and geometric reconstruction techniques, these BIM-based platforms enhance model fidelity

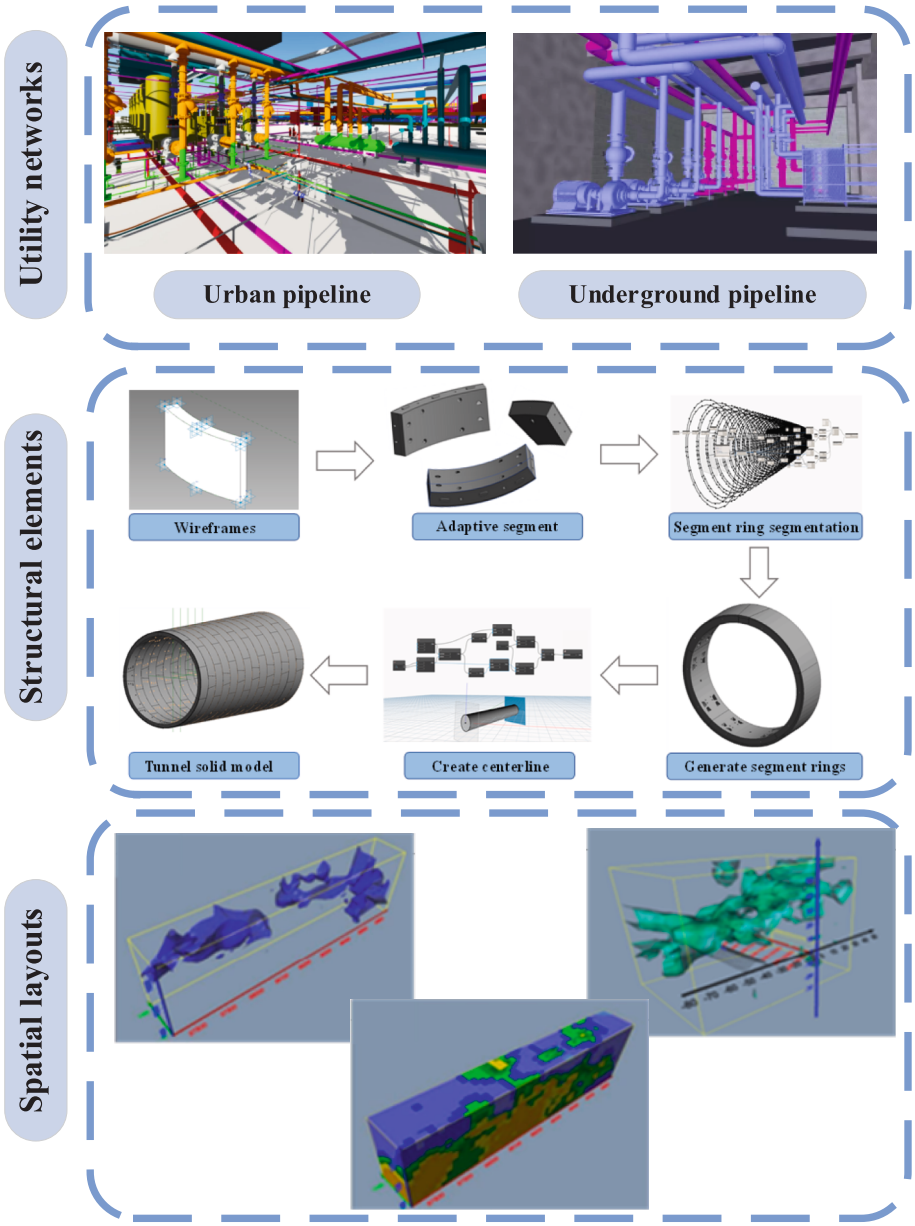

Fig. 2. The application of BIM models (Li et al., 2026).

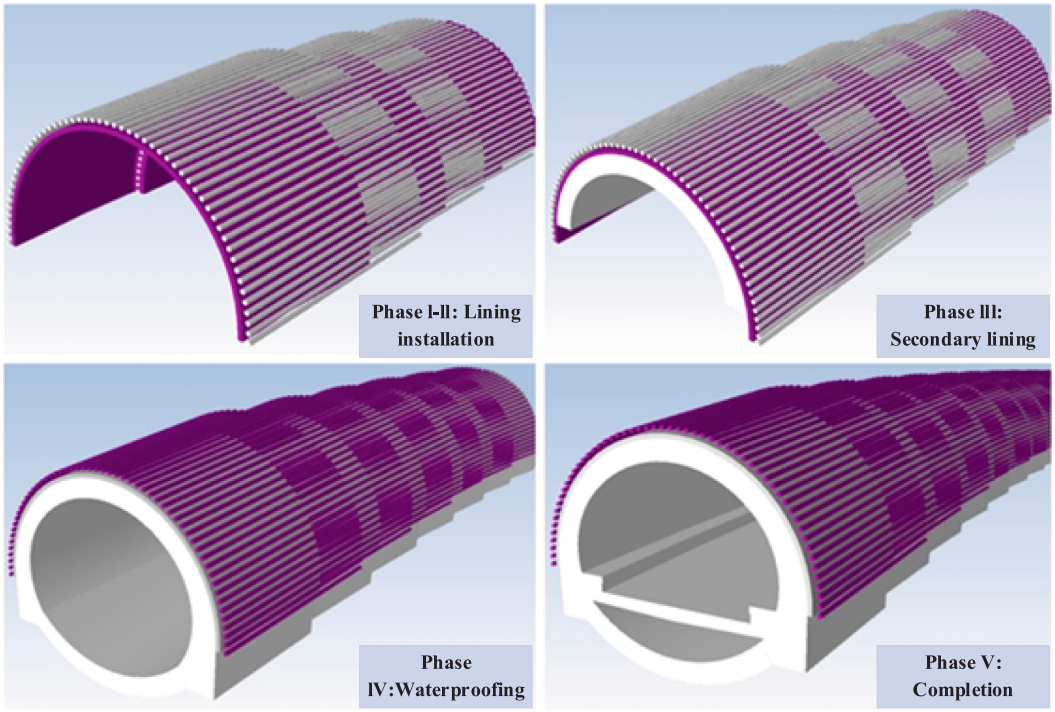

Fig. 3. SyncroPro: 4D model of the railway tunnel construction process (He et al., 2025).

and support detailed analysis of structural responses under diverse geological conditions (Bui et al., 2024; Fabozzi et al., 2021; Waleed et al., 2024).

Despite recent progress, structural visualization technologies in tunnel engineering still face notable limitations. Most existing tools operate in silos, disconnected from structural analysis, real-time monitoring, and O&M platforms, which feedback-driven decision systems. Compared to BIM-FEM integration in buildings or digital twin frameworks in highway infrastructure, tunnel-specific solutions often lack interoperability and system-level coordination. Moreover, current models are predominantly static, failing to capture the evolving mechanical behavior and risk propagation under dynamic geological or loading conditions. The absence of standardized data interfaces across design, construction, and maintenance phases further restricts model reusability and lifecycle integration. Future research should focus on coupling structural visualization with real-time sensing, AI-enhanced analytics, and adaptive simulation modules. Establishing open, extensible data protocols will be key to bridging fragmented platforms and enabling automated model updates and decision support. The integration of digital twin paradigms with machine learning also holds significant promise for improving the accuracy, robustness, and interpretability of tunnel design and risk management under uncertainty.

2.1.2. Advanced geological prediction

Advanced geological prediction during the design phase leverages modern exploration techniques such as ground-penetrating radar (Xin, 2017), acoustic wave surveys (Liu et al., 2023a), seismic reflection imaging (Aoki et al., 2007) and borehole drilling (Fang et al., 2021). These methods yield critical subsurface data, including stratigraphy, geomechanically properties, hydrogeological conditions, and geological discontinuities. Such information supports the identification of weak zones, prediction of adverse geological features ahead of the excavation face, assessment of ground deformation trends, and estimation of seismic response behaviors. By integrating multi-source datasets, accurate 3D geological models can be constructed, forming the basis for selecting appropriate tunneling methods, designing support systems, and mitigating geological risks, as illustrated in Fig. 4. To enhance geological section prediction in scenarios with sparse borehole data, Qiu (Qiu et al., 2024) proposed a multi-spatial receptive field (MSRF) XGBoost model in Fig. 5. This model combines classification and recognition modules with multi-scale spatial feature extraction and automated model selection mechanisms, thereby improving robustness in complex geological settings.

In addition to data-driven prediction, a range of 3D geological modeling approaches have been explored to address limited and heterogeneous subsurface data. Cooperative kriging methods fuse multi-source inputs to improve the representation of complex formations and enable dynamic model updates (Zhang and Zhu, 2018). Data-driven frameworks such as IC-XGBoost3D combine sparse borehole records with 2D geological images to construct 3D models while quantifying uncertainty (Shi and Wang, 2022). Multiscale modeling techniques further integrate field reconnaissance, drilling, and geological forecasting for construction support (Xiong et al., 2018). Improved coupled Markov chain (CMC) methods, as shown in Fig. 6, enhance lateral stratigraphic simulation by integrating Walther's Law with vertical and intermediate transition matrices. This approach reduces horizontal transition probability errors to under 3 % and remains robust across varying borehole layouts. (Zhang et al., 2022a).

Collectively, these approaches address major challenges in 3D geological modeling by improving prediction accuracy, enabling adaptive construction-oriented modeling, and enhancing the fidelity of complex stratigraphic simulations. Nevertheless, advanced prediction methods still face critical limitations, including restricted generalizability in heterogeneous geology, high computational demands, and strong reliance on data quality and coverage. The intrinsic complexity of subsurface conditions continues to hinder accurate site reconstruction. Future research should prioritize improving model scalability and robustness, coupling with real-time sensing, and advancing physics-informed machine learning to achieve reliable and practical predictions under complex and dynamic underground conditions.

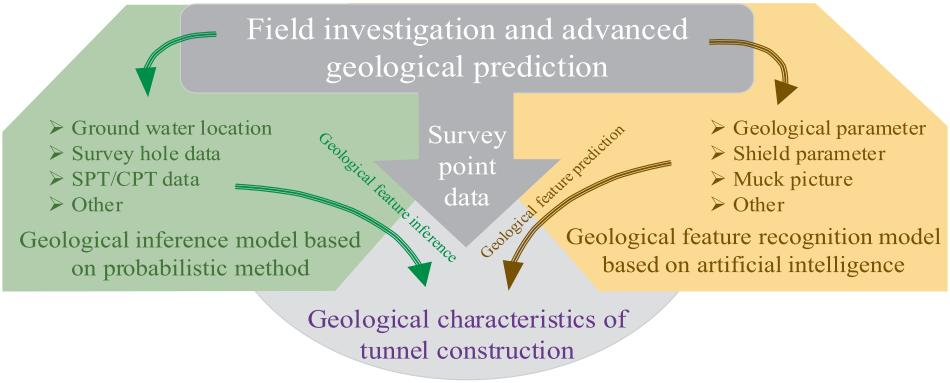

Fig. 4. Summary of geological exploration methods.

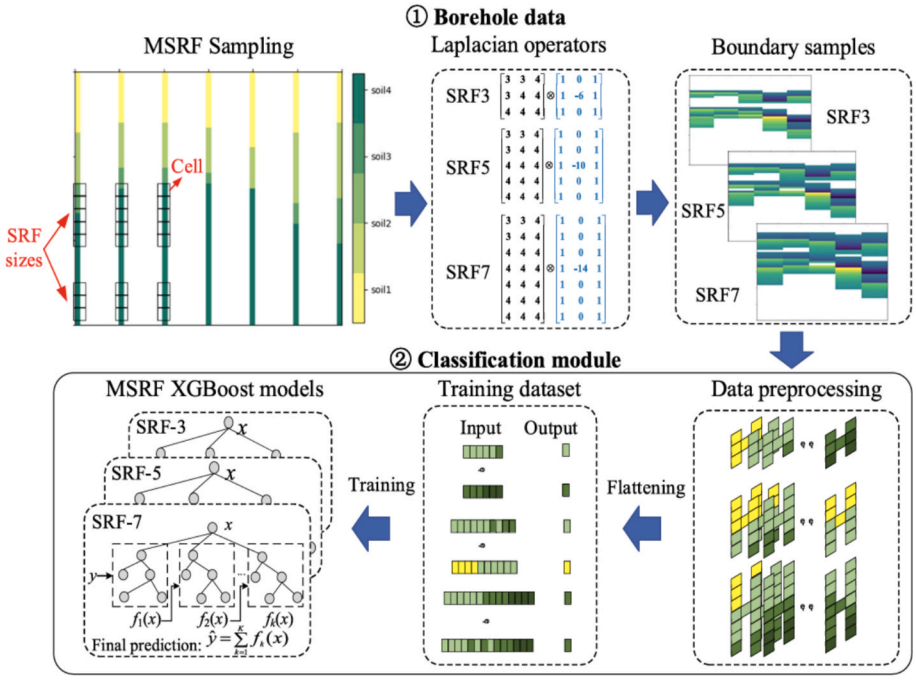

Fig. 5. Illustration on classification module of MSRF XGBoost method (Qiu et al., 2024).

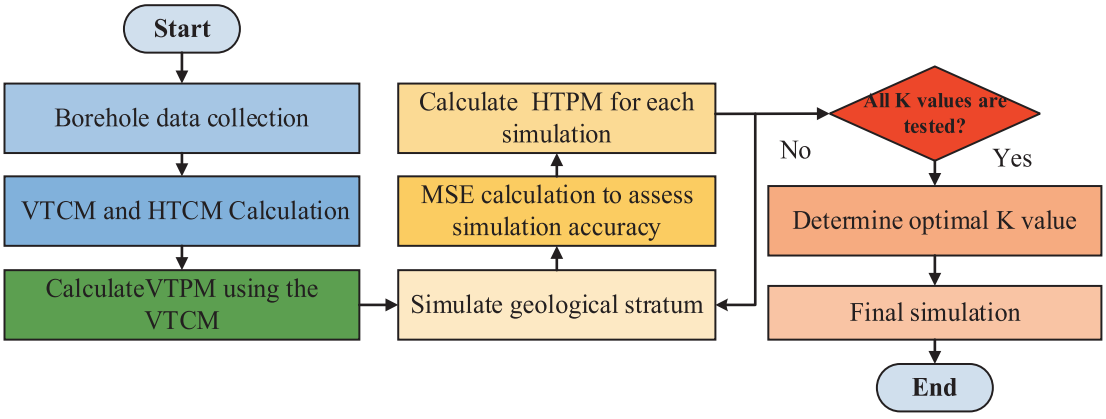

Fig. 6. Improved coupled Markov chain model (Zhang et al., 2022a).

2.2. *Application of key technologies in the tunnel construction phase*

The construction phase of tunneling critically depends on the integration of advanced technologies to ensure safety, efficiency, and quality. Real-time data acquisition provides timely insights into geological and environmental conditions, while risk identification

techniques enable early warning and preventive control of hazards. Concurrently, parameter optimization enhances operational efficiency and strengthens quality management. The combined deployment of these technologies establishes a scientific basis for tunneling in complex geological settings and underpins the successful delivery of projects. To comprehensively capture the state-of-the-art technological applications during the tunnel construction phase, this section synthesizes representative studies retrieved using keywords such as 'real-time data acquisition', 'construction risk assessment', 'parametric prediction', 'digital twin for construction', and 'intelligent tunneling control systems' across the Web of Science.

2.2.1. Data acquisition

Real-time data acquisition provides the foundation for intelligent decision-making and automated control in tunnel construction. Key data streams include: (i) environmental parameters such as temperature, humidity, gas concentration, and particulate matter, which ensure safe working conditions in enclosed underground settings;(Jha and Tukkaraja, 2020; Muñiz et al., 2023; Weng et al., 2022) (ii) structural response data (displacement, stress, strain) that reflect the stability of surrounding rock and tunnel linings during excavation and initial support (Ling et al., 2024; Zhou et al., 2020); and (iii) progress-related indicators such as construction stage, equipment runtime, and material consumption, which enable schedule tracking and resource management.(Xu et al., 2023a) Multi-source sensing systems integrating these datasets have already been validated in metro and highway tunnel projects, demonstrating effectiveness in early warning, real-time risk assessment, and adaptive control. As shown in Fig. 7, such systems form the backbone of intelligent construction monitoring platforms and provide transferable methodologies adaptable to other large-scale underground engineering projects.

A critical element of data acquisition in tunnel construction is the real-time monitoring of shield tunnelling parameters such as propulsion force, cutterhead speed, torque, and chamber pressure. These parameters, captured by embedded sensor arrays, are vital for detecting abnormal conditions, enabling adaptive control, and mitigating risks including ground settlement and segmental deformation. Recent advances highlight three development trends: (i) enhanced sensor precision, durability, and fault tolerance for reliable monitoring in harsh underground environments; (ii) the adoption of edge computing and 5G communication to accelerate data transmission and support real-time early warning and closed-loop control; and (iii) the integration of AI, machine learning, and big data analytics to optimize tunnelling parameters and guide operational decisions. These technologies have been successfully applied in metro and cross-river shield tunnel projects, demonstrating their practical value and scalability across different geological settings. A summary of representative developments is provided in Table 1.

Building on the multi-source data acquisition discussed above, effective data utilization requires further processing and optimization through integrated algorithmic frameworks. These algorithms encompass data cleaning, noise filtering, and data fusion technologies, which collectively enhance data accuracy, consistency, and reliability. Additionally, by combining advanced technologies such as machine learning and artificial intelligence with both historical and temporal data, it becomes possible to dynamically predict shield machine behavior and ground responses, optimize tunnelling parameters such as propulsion force and cutterhead torque, and support adaptive control to ensure construction efficiency and safety. Representative application cases of these integrated technologies are summarized in Table 2.

Despite recent advances, sensor systems for tunnel construction still encounter significant challenges under complex and dynamic underground conditions. Unlike static structures, tunneling involves stress

Table 1
Development trends of tunnel sensor technologies.

Category	Representative Work	Key Features
High Precision	(Liu et al., 2020)	Introduced TMR-based non-contact displacement sensors and micro-displacement sensors with nanometer-level resolution;
	(Ren et al., 2021)	Quartz resonator-based force sensors expand accuracy across a wide measurement range.
	(Wang et al., 2022)	
Self-Diagnosis	(Park et al., 2008) (Park et al., 2009) (Jiang and Luo, 2024)	Combined electromechanical impedance models with defect influence analysis; developed self-diagnostic systems for piezoelectric elements; applied GCN to realize multi-sensor autonomous health evaluation.
Low Energy Consumption	(Gao et al., 2022) (Lin et al., 2010) (Al Azim et al., 2017)	Applied MEMS sensors for real-time monitoring with low power use; proposed star-topology wireless sensor networks to reduce energy and cost; developed low-voltage circuits using magnetic skyrmions.
Multiple Integration	(Li et al., 2023a) (Huang et al., 2019) (Lu et al., 2022)	Designed PM-FBG-based multi-parameter sensors for simultaneous monitoring; realized concurrent strain, temperature, and pressure detection; achieved refractive index sensing in complex tunnel environments.

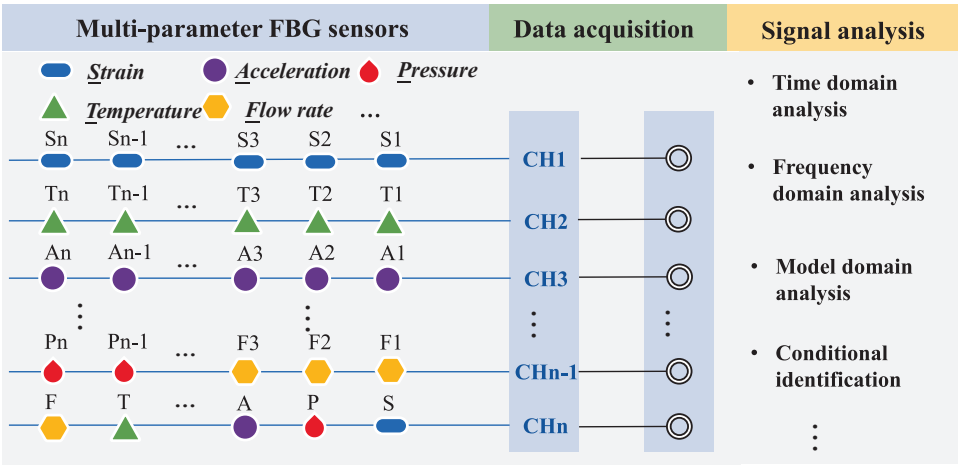

Fig. 7. Multi-parameter dynamical measuring system using FBG sensors (Huang et al., 2019).

Table 2
Shield parameter prediction technology.

Method	Main point	Advantage	Representative Case Study
GA (Genetic Algorithm) (Fattahi and Jiryaee, 2022)	Global optimization via evolutionary operators	Strong global search, flexible encoding	Bangkok Subway settlement reliability analysis: GA + Neural Network + FORM/SORM achieved R = 0.99, RMSE = 0.0117; better accuracy than PSO, runtime ≈ 5 min
PSO (Particle Swarm Optimization) (Hu et al., 2023)	Swarm-based parameter optimization	Fast convergence, simple implementation	Same Bangkok Subway study: PSO obtained slightly lower accuracy but similar runtime (~5 min). Applied in pipe-jacking tunnels with PSO–BP model for settlement prediction.
ACO (Ant Colony Optimization) (Afradi et al., 2020)	Path-finding and combinatorial optimization	Robust for discrete problems, suitable for routing	Sabzkooh water conveyance tunnel: TBM penetration rate prediction, performance comparable to PSO but more parameter-sensitive.
GWO (Grey Wolf Optimizer) (Abdelkader et al., 2022)	Hierarchical leadership-based optimization	Balanced exploration & exploitation; fewer parameters	Highway tunnel deterioration modeling: GWO–Gaussian Process Regression hybrid, improved convergence with fewer iterations.
GWO-FW-MKL-SVR (Yang et al., 2022)	Predicts TBM penetration rate using multi-source data fusion and biogeographic models	GWO optimization enhances model accuracy and stability	Applied to a large-scale water conveyance tunnel project; optimization objective: penetration rate prediction; result: improved RMSE by 12 % vs. baseline SVR; efficiency: fewer iterations required due to GWO tuning.
BAS-BPSO-XGBoost (Wang et al., 2023b)	Introduces Beetle Antennae Search into particle swarm to optimize XGBoost hyperparameters	Enhances prediction precision and search capability compared to standard PSO	Shenzhen tunnel dataset; optimization objective: TBM thrust and penetration prediction; result: MAE reduced by 15 % vs. PSO-XGBoost; efficiency: moderate runtime increases due to BAS integration.
Integrated Regression (Shi et al., 2023)	Combines BAS and PSO for subspace regression	Improves robustness and accuracy by adapting models to subspace data characteristics	Field dataset from Guangzhou metro tunneling; optimization objective: multi-parameter regression (thrust, torque, penetration); result: improved robustness across heterogeneous strata; efficiency: moderate computation time with stable convergence.

redistribution, groundwater migration, and strong mechanical vibrations, all of which undermine the accuracy, durability, and calibration stability of sensing devices. Fiber Bragg Grating (FBG) sensors, for example, offer high sensitivity but remain vulnerable to signal interference and degradation in high-moisture environments, while MEMS sensors often suffer from long-term durability issues and calibration drift. In addition, latency in data transmission and limited energy supply in deep-buried environments restrict both real-time monitoring and long-term deployment. From an engineering perspective, inconsistent sensor deployment protocols across different projects further hinder data interoperability and reduce the transferability of monitoring models. Addressing these limitations requires sensor technologies that combine robustness and environmental adaptability, supported by energy-efficient or self-powered nodes to ensure sustainability. Moreover, the establishment of standardized deployment and calibration guidelines across tunneling projects will be essential for improving data comparability, facilitating model reuse, and ultimately enabling scalable intelligent monitoring systems for complex geotechnical environments.

2.2.2. *Risk assessment*

Building intelligent monitoring and risk assessment is a core component of shield tunnel construction, systematically evaluating the probability and consequences of potential risks to support hazard control and risk-informed decision-making. Given the dynamic and uncertain nature of underground conditions, robust frameworks improve the reliability of construction strategies across varying scenarios. In practice, tunnel projects commonly classify risks into four categories: project-inherent, geological, environmental, as shown in Fig. 8, which have proven valuable for identifying critical risk sources and prioritizing mitigation measures.

To support effective risk assessment in shield tunnel construction, diverse approaches have been applied to identify, evaluate, and manage construction-related risks. Common methods include expert surveys, fault tree analysis (FTA), event tree analysis (ETA), and machine learning models, which together enable both qualitative and quantitative evaluation of hazards and guide mitigation strategies. According to the 'Code for risk management of underground works in urban rail transit' (Ministry of Housing and Urban-Rural Development of the People's Republic of China, 2011), assessment methodologies in rail transit construction are generally categorized into qualitative, quantitative, and semi-quantitative types, as summarized in Table 3.

The tunnel construction phase is exposed to diverse and dynamic risks, including differential settlement, water and mud inrush, tunnel face instability, and disturbances from adjacent activities. These hazards are marked by high uncertainty, hidden evolution, and sudden onset, which challenge traditional empirical judgment and static monitoring.

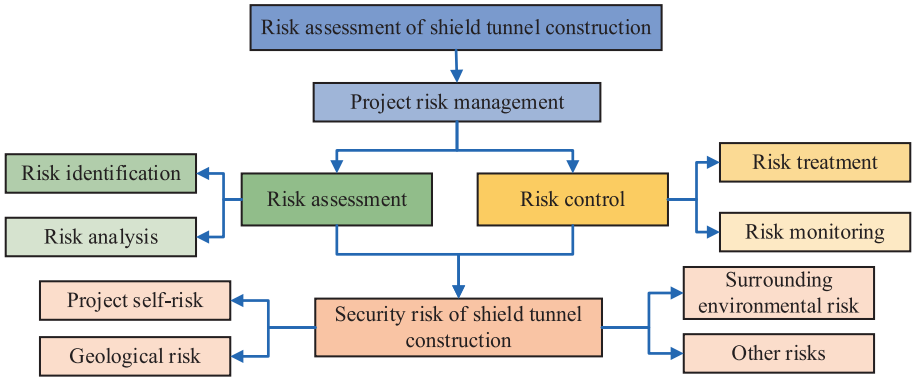

Fig. 8. Shield tunnel risk assessment.

Table 3
Risk assessment method.

Qualitative analysis method	Quantitative analysis method	Comprehensive analysis method
Expert review + CIM + risk database (Huang et al., 2006)	Stacked deep learning + multi-source data + simulation (Zhou et al., 2023a)	Fuzzy set analysis (FSPA) + fuzzy C-means (Zhou et al., 2024b)
BIM + expert system (B-RIES) (Zhang et al., 2016)	Bayesian network for gas pipeline dynamics (Fang et al., 2019)	GCN + uncertain parameters (Xue et al., 2024b)
AHP–EGM approach for station-level risk (Liu et al., 2021)	Stochastic geological modeling + MC + GUI (Zhang et al., 2021a)	Fuzzy set theory + DBN for full-process risk (Wang et al., 2023a)

Recent advances in AI, IoT, big data analytics, and digital twin systems have transformed risk identification, assessment, and management practices in tunneling. The following section reviews the development and application of these technologies in addressing four critical categories of construction risks.

(1) Uneven settlement of soil.

Uneven soil settlement during tunnel construction threatens both construction safety and the stability of overlying infrastructure. Recent advances show that machine learning (ML) techniques significantly outperform empirical and statistical models in predicting settlement patterns. Supervised algorithms such as RF, XGBoost, SVM, and LSTM have been applied to shield tunnelling projects, effectively combining operational parameters, geotechnical data, and monitoring records to improve prediction accuracy and adaptability (Wang et al., 2024d; Ye et al., 2022). Hybrid soft computing approaches that integrate fuzzy logic with neural networks further capture the nonlinear and uncertain nature of soil-structure interactions in EPB tunnelling (Zhang et al., 2024b; Zhang et al., 2021c). In addition, probabilistic risk evaluation frameworks have been introduced to incorporate data uncertainty and construction variability into settlement risk metrics, supporting more reliable decision-making under uncertainty (Chen et al., 2024b).

Complementing predictive models, remote sensing and multi-source monitoring techniques enhance spatial and temporal coverage. InSAR using Sentinel-1 imagery has been successfully employed in dense urban areas to detect tunnelling-induced deformation (Liu et al., 2023b). At the local scale, distributed optical fiber sensors such as BOTDA, integrated into IoT frameworks, enable continuous high-frequency monitoring and early warning of anomalous settlement. Furthermore, BIM-GIS integration provides intuitive spatial visualization of risk zones and improves information sharing across project stakeholders. Collectively, these developments illustrate a shift from reactive monitoring to predictive and proactive settlement management, with demonstrated value in metro and urban tunnel projects.

(2) Tunnel water and mud inrush.

Water and mud inrush represent one of the most severe hazards in shield tunnelling, typically triggered by face instability or sudden groundwater intrusion. Intelligent technologies have greatly advanced risk identification and early warning. Machine learning models, including ensemble methods and SVM, have shown high predictive accuracy under complex geological conditions (Mahmoodzadeh et al., 2021). Multi-criteria decision-making frameworks such as FAHP-I-TOPSIS and uncertainty theory provide structured evaluation of key indicators including permeability, water pressure, and geological anomalies (Dou et al., 2023; Kong and Zhang, 2024; Zhao et al., 2024b), supporting both qualitative and quantitative decision-making, as shown in Fig. 9.

To better capture uncertainty, multi-source geophysical data combined with improved Dempster-Shafer evidence theory enables probabilistic identification of potential inrush zones (Ding et al., 2022). Bayesian networks and fuzzy inference systems have further been

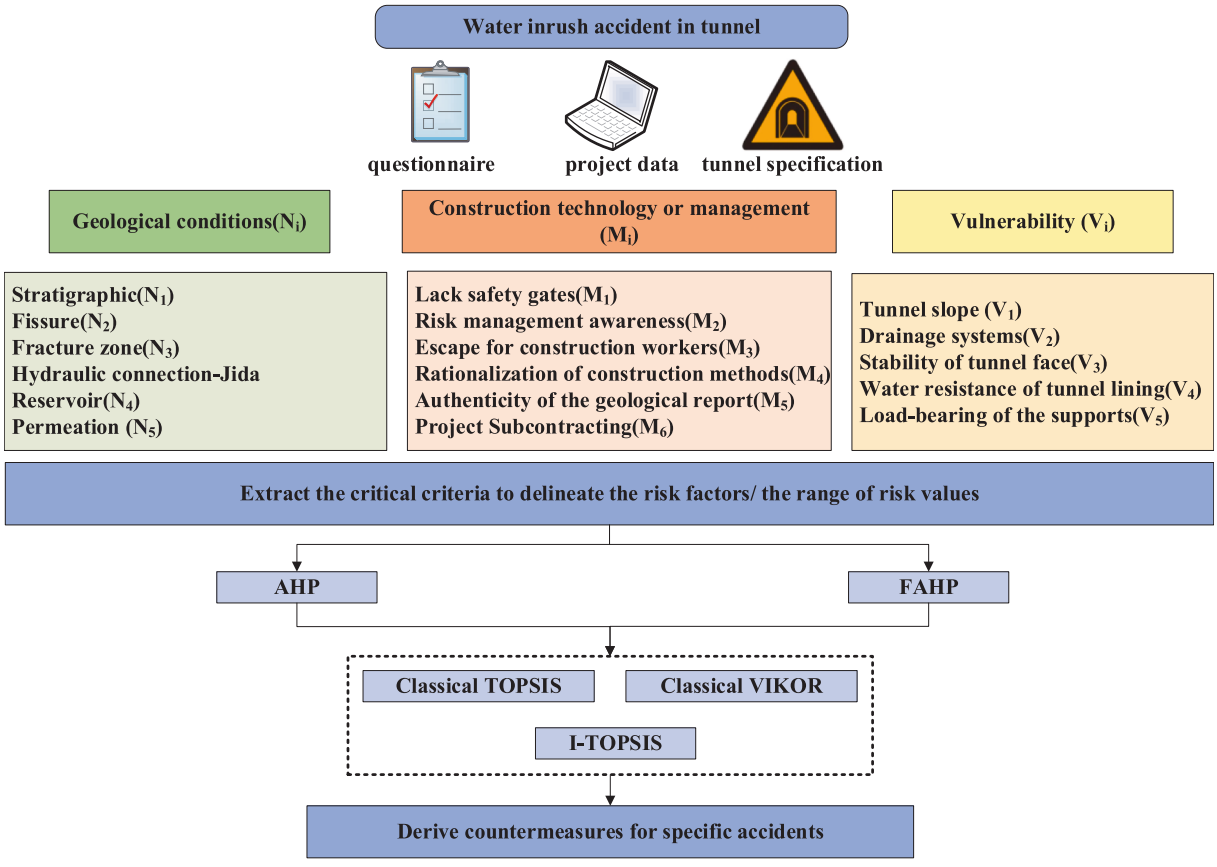

Fig. 9. Indicator system diagram of the assessment structure (Kong and Zhang, 2024).

applied to assess fault-related and subsea inrush scenarios, effectively modeling complex dependencies with sparse datasets (Kong et al., 2025; Song et al., 2021; Sun et al., 2022). More recently, digital twin platforms allow dynamic simulation of geological conditions and hazard evolution, enhancing predictive accuracy in high-risk contexts such as high-temperature water inrushes (He et al., 2024). Additionally, heterogeneous data fusion that integrates sensor outputs, survey results, and expert knowledge improves both sensitivity and robustness of early warning (Zhang et al., 2025c). Collectively, these advances mark a transition from reactive hazard response to proactive, data-driven risk management, with demonstrated applicability in metro and subsea tunnel projects.

(3) Collapse of the tunnel.

Tunnel face collapse is a persistent challenge in underground construction, especially in karst zones, faulted strata, and weak rock masses. Recent studies demonstrate the value of intelligent modeling and probabilistic evaluation for risk identification and control. For example, Yang et al. (Yang et al., 2023b) developed a hybrid random forest-based model incorporating metaheuristic optimization to forecast shield-induced ground settlement in Singapore, achieving high accuracy across three real EPB projects. Similarly, Wang et al. (Wang et al., 2024d) employed a PSO–BP neural network calibrated with monitoring data from a metro project, demonstrating strong agreement with field-observed settlement profiles. These studies affirm the practical potential of data-driven approaches in early warning and control of tunnel face instability.

Advanced algorithms further support early warning and probabilistic assessment. PSO-optimized LSTM models capture nonlinear settlement trends near tunnel portals (Sun et al., 2025b), and SVR combined with Monte Carlo simulation quantifies collapse probability under uncertainty (Meng et al., 2023). These data-driven approaches improve risk quantification based on construction monitoring data and geological features, facilitating targeted mitigation. Complementary methods such as cloud models, Bayesian networks with self-learning capacity, and entropy-weighted clustering (Ju et al., 2024; Sun et al., 2024) allow dynamic refinement of risk classification, while hybrid frameworks incorporating extension theory have been applied in karst terrains to address spatial variability (Zhang et al., 2022b), as shown in Fig. 10.

Expert knowledge systems, including analytical hierarchy process (AHP) and Delphi methods, have been increasingly integrated with quantitative models to enhance interpretability and practical adoption, and information fusion techniques such as enhanced Dempster-Shafer theory improve robustness under ambiguous or sparse data (Huang et al., 2023; Kim et al., 2022). Collectively, these developments signify a critical shift from traditional experience-based evaluation toward data-integrated, adaptive, and interpretable frameworks for tunnel collapse risk prediction, ultimately enabling more resilient and responsive construction practices.

(4) Disturbance of adjacent buildings.

Shield tunnelling in dense urban areas can induce deformation and damage in adjacent buildings and underground structures, making proximity disturbance a critical external risk. To address the limitations of traditional assessment methods, which often overlook dynamic weight variation and uncertainty, recent studies have introduced advanced probabilistic and fuzzy frameworks. For example, risk assessment models combining variable-weight theory with cloud modeling flexibly capture fuzziness and randomness in disturbance scenarios (Guo et al., 2024), while improved FAHP approaches integrating game-theoretic weighting and nonlinear cloud models (Guo et al., 2025), enhance the identification of dominant risk factors, as shown in Fig. 11. The developed model was applied to assess shield tunneling risks near buildings along Changsha Metro Line 4, identifying 'medium' to 'high' risk levels. The results aligned well with actual site conditions, validating the model's reliability.

In addition, hybrid models have also been developed to support dynamic and probabilistic evaluations. A representative example is the integration of the Trapezium Cloud Model (TCM) with Bayesian Networks (BN), where TCM outputs are transformed into key risk indicators

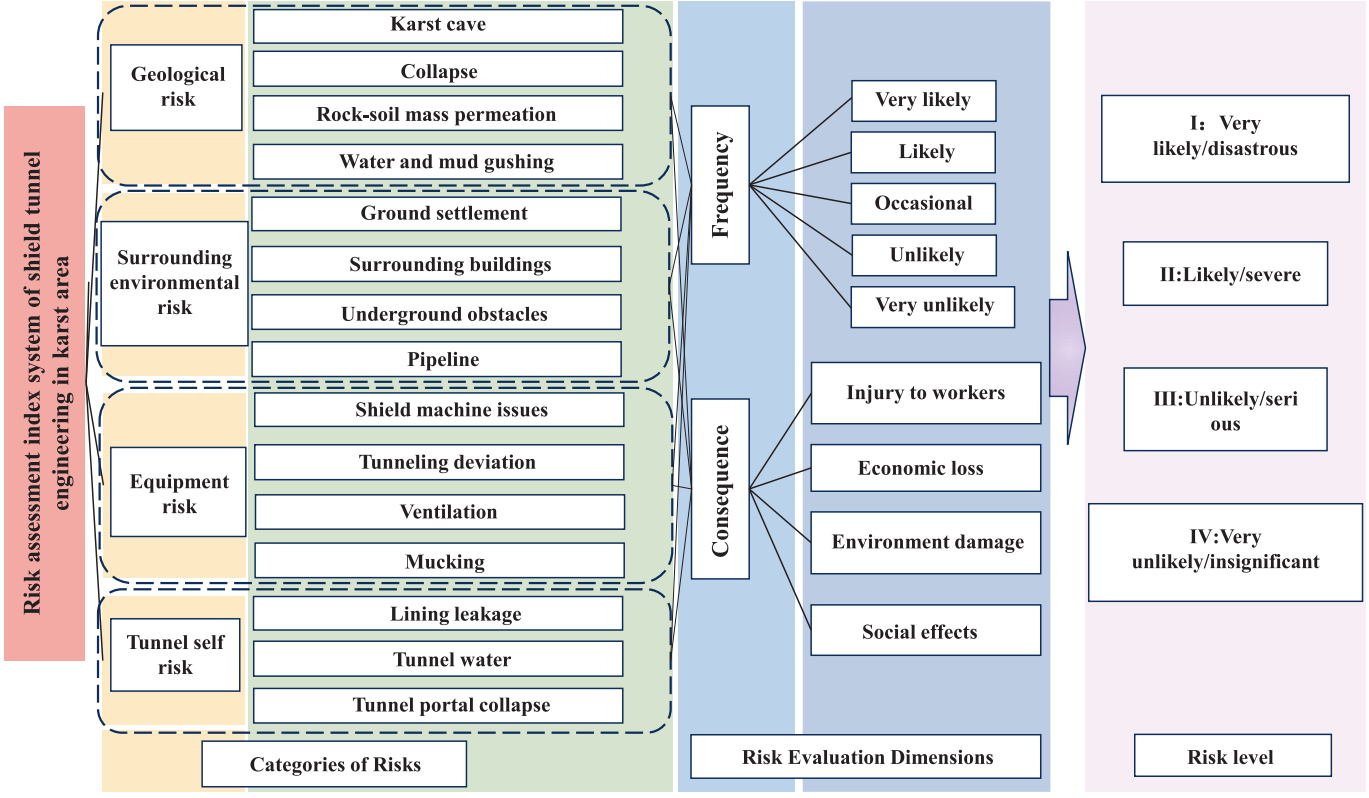

Fig. 10. Risk index system for shield tunneling projects (Sun et al., 2024).

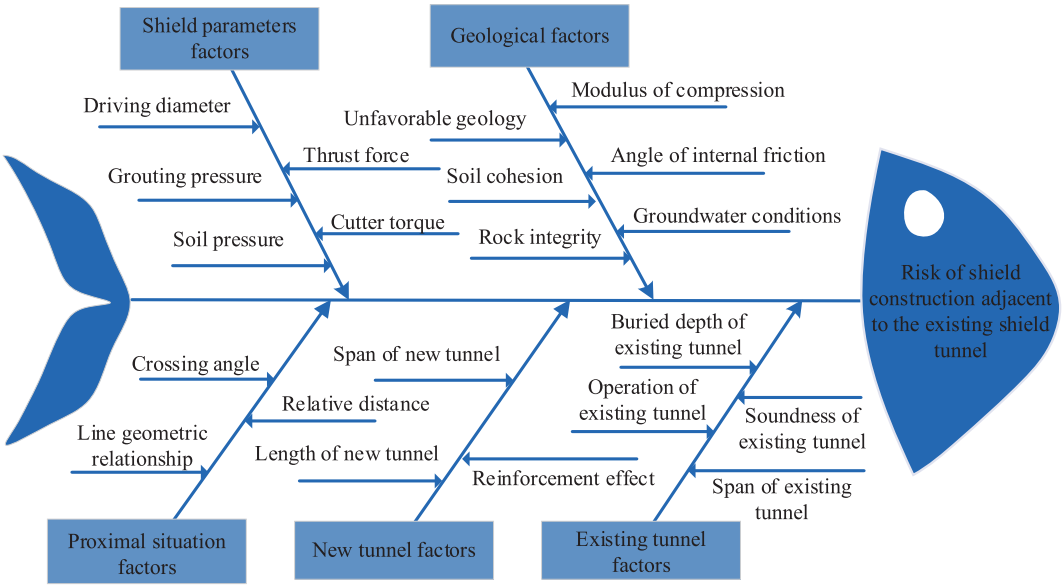

Fig. 11. Diagram of shied construction adjacent to the existing shield tunnel (Guo et al., 2025).

to enable forward reasoning, sensitivity analysis, and diagnostic evaluation (Chen et al., 2024a). The predicted safety risk levels at monitoring points using this trapezoidal cloud-BN model showed good agreement with on-site assessments, with the risk probabilities at level III for points Q2, Q4, Q6, and Q8 reaching 47.3 %, 37.6 %, 42.9 %, and 42.2 %, respectively.

The preceding review highlighted advances in assessing four major risks in shield tunnelling: uneven soil settlement, water and mud inrush, tunnel face collapse, and disturbance to adjacent structures. Despite advances in machine learning, probabilistic evaluation, and multi-source monitoring, key challenges remain. Current models struggle with heterogeneous data integration, lack dynamic adaptability to evolving risk states, and suffer from inconsistent indicators across projects. Future work should focus on real-time data fusion with AI and causal inference, unified and adaptive assessment frameworks, and cross-domain data integration to strengthen robustness and decision support for intelligent tunnel platforms.

2.2.3. *Parametric prediction*

Accurate forecasting of key operational parameters is central to intelligent shield tunneling, ensuring construction continuity and structural safety. The tunneling process is affected by geological conditions, equipment performance, this section focuses on three high-impact tasks: (i) prediction of cutter wear, supporting predictive maintenance and tool-change planning via learning-based and physics-informed surrogates using torque-power-penetration signatures; (ii) prediction of shield attitude control, enabling feed corrections through sequence models that map ground conditions and operational inputs to deviation risk; and (iii) prediction of shield muck identification, classifying face/soil types and conditioning needs from sensor fusion to stabilize excavation. Each sub-section summarizes representative methods, required data, engineering maturity, and scalability considerations.

(1) Prediction of shield cutter plate wearing.

Cutter wear is a critical parameter in shield tunnelling, as excessive degradation leads to downtime, higher maintenance costs, and reduced excavation efficiency. Accurate prediction supports optimized tool-change planning and continuous progress. With advances in machine learning, models such as LSTM (Wang et al., 2024b), gene expression programming (GEP) (Tao et al., 2022), ensemble learning, and attention-based architectures (Zhang et al., 2023b) have been applied using torque, thrust, penetration rate, and geological indices, demonstrating strong predictive performance across diverse strata (Li et al., 2023b; Mo et al., 2024; Shin et al., 2024; Wang et al., 2025a), as shown in Fig. 12. Hybrid approaches that integrate physical indicators such as the Cerchar Abrasivity Index (Sun et al., 2023) and energy-based features (Wang et al., 2012) enhance generalization, while frameworks combining sensor data with mechanical models improve robustness under complex geological conditions (Elbaz et al., 2021). Kwon et al. (Kwon et al., 2025) have developed an interpretable XGBoost model with data augmentation to predict abnormal cutter wear under complex mixed ground conditions. Trained on 65 CHI reports, the model achieved a recall of 0.80 and an F1-score of 0.808. Through model interpretation, key influencing factors, such as chamber pressure, cutter installation radius, and torque were identified. The incorporation of equivalent elastic modulus and class-balancing strategies effectively addressed geological variability and data imbalance, providing valuable insights into the mechanisms of abnormal wear.

However, most models remain project-specific, with limited transferability across different strata, and face challenges from sensor noise, data heterogeneity, and computational demands. Moreover, the black-box nature of deep learning restricts acceptance in field operations. Future research should emphasize cross-project benchmark datasets, multi-source fusion, and hybrid data-mechanics frameworks to enable robust, interpretable, and scalable cutter wear prediction, as already piloted in metro and cross-river tunnel projects.

(2) Prediction of shield attitude control.

Shield attitude control refers to predicting and adjusting the machine's spatial orientation during excavation, which is critical for alignment accuracy and tunnel geometry. Traditional deterministic methods lack adaptability under complex conditions, prompting the adoption of data-driven approaches that combine historical excavation records with real-time sensor data. Hybrid deep learning models with attention mechanisms have proven effective in capturing nonlinear relationships between geology, operational parameters, and posture (Dai et al., 2023; Kang et al., 2023; Wang et al., 2025b), while Bayesian-based models such as Bayesian-LGBM improve robustness through explicit uncertainty quantification (Chen et al., 2023a; Wang et al., 2024c). Dimensionality reduction techniques like PCA further stabilize training in high-dimensional tunneling datasets (Zhang et al., 2025b).

For real-time control, multi-source fusion frameworks integrating sensor feedback, geological data, and numerical simulations have been applied in large-diameter and slurry TBMs, achieving more adaptive and reliable attitude correction (Fu et al., 2024b; Tian et al., 2024; Wu et al., 2024b). Emerging multi-agent deep reinforcement learning (SCA-

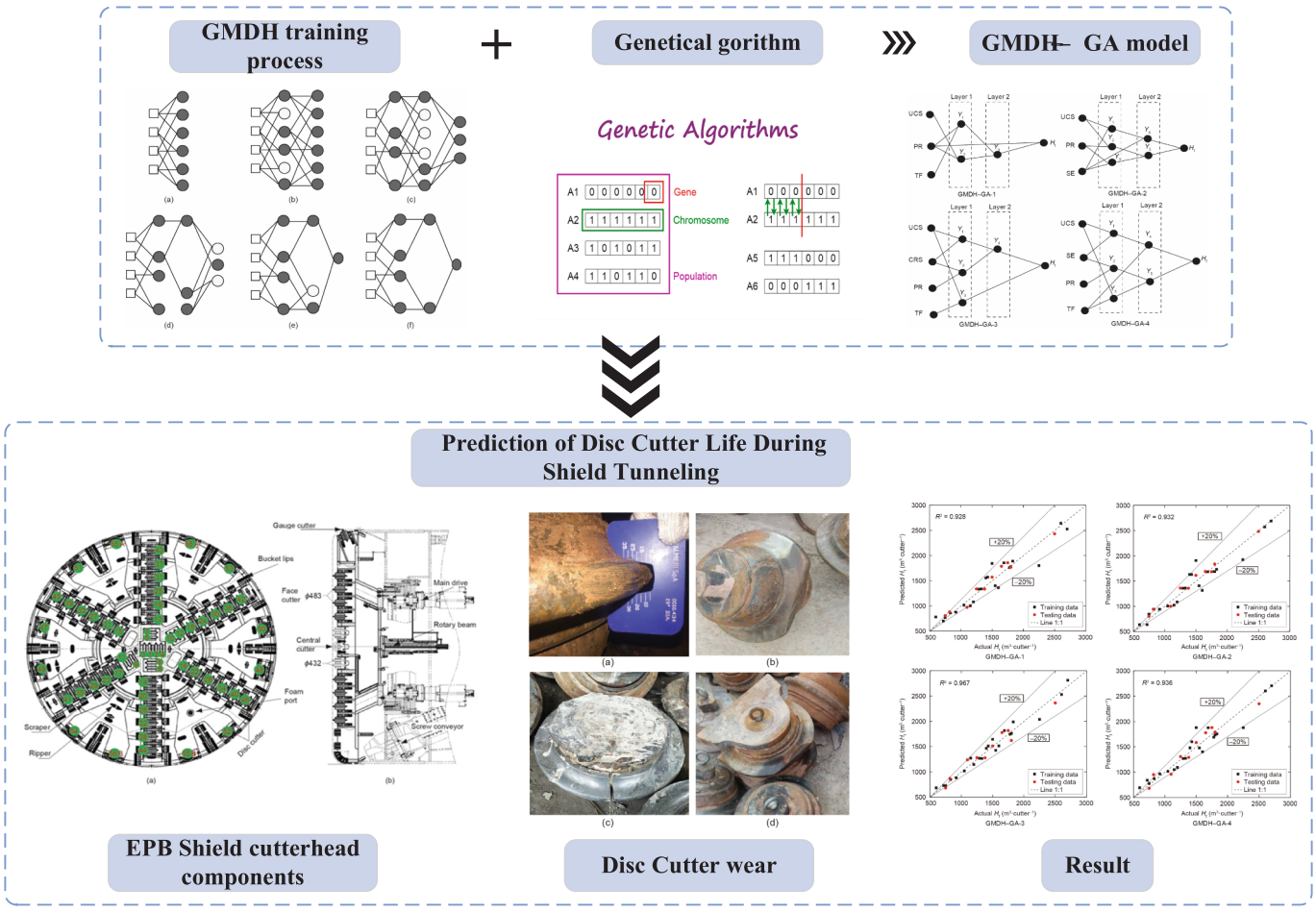

Fig. 12. General framework for the prediction of disc cutter life (Shin et al., 2024).

MADRL) enables autonomous decision-making and multi-objective optimization for shield posture (Fu et al., 2024a; Xu et al., 2024b), as shown in Fig. 13. Despite these advances, current models remain project-specific, with limited transferability across tunneling scenarios, underutilized spatiotemporal dependencies, and insufficient integration of physics-informed constraints. Latency between prediction outputs and control actions also restricts responsiveness (Fu et al., 2023; Sun et al., 2025a). Future work should focus on transferable, interpretable,

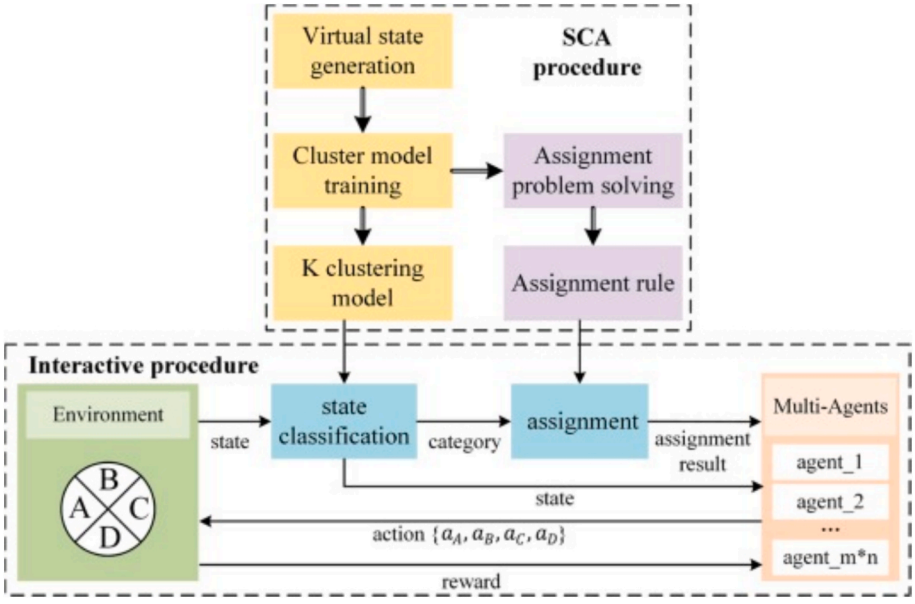

Fig. 13. The SCA-MADRL framework for shield attitude adaptive control (Fu et al., 2023).

and hybrid physics-data frameworks to support robust, real-time shield attitude control, as recent pilot projects have begun to demonstrate.

(3) Prediction of shield muck identification

Muck properties provide critical feedback on excavation conditions and subsurface geology. Accurate identification supports adaptive control strategies and cutter management. Recent methods combine tunneling parameters with muck physical features for real-time classification of face rock mass (Huang et al., 2025). Image-based multi-scale analysis enables precise evaluation of particle size and distribution (Baek and Choi, 2025), while AI-assisted spectroscopy techniques such as laser-induced breakdown spectroscopy (LIBS) offer quantitative insights into mineral composition, including quartz content, indirectly reflecting lithological properties (Wang et al., 2024g).

Deep learning approaches further enhance classification accuracy. Ensemble frameworks such as EMNet integrate shield operation data and muck features for geological detection (Fu et al., 2025). CNN-based architectures including enhanced Faster R-CNN, have been applied to classify muck type and estimate plasticity in soft soils (Zhang et al., 2023a; Li et al., 2022), as shown in Fig. 14.

Synthesizing the above on cutter wear, shield attitude, and muck identification, parametric prediction has moved from heuristic rules to data-driven models and has shown promise in field pilots. However, deployment at scale remains constrained by heavy labeling demands, distribution shifts across strata or TBM types, sensor noise and heterogeneity, and latency/compute limits for real-time control. Most models are task-specific, with limited transferability and weak uncertainty quantification, and the lack of cross-project benchmarks hinders objective comparison and reuse (Yan et al., 2023). To address these challenges, future research should explore these four priorities for engineering-ready systems: (i) transfer learning techniques that adapt pretrained models from data-rich projects to new tunneling contexts with limited labeled data. (ii) physics-informed machine learning that embeds geotechnical knowledge (constitutive models, boundary conditions) to improve interpretability and physical consistency. (iii) cross-project benchmark datasets covering diverse geological, structural, and operational conditions to enhance robustness and support generalization. (iv) multi-source sensing data fusion integrating advanced monitoring techniques such as remote sensing, ground-penetrating radar, and fiber Bragg grating-based systems, to provide comprehensive excavation feedback and improve prediction reliability.

2.2.4. *Intelligent prediction in drill-and-blast tunnel construction*

Although shield tunnelling dominates intelligent construction, the drill-and-blast method remains indispensable in mountainous and geologically complex regions. Integrating intelligent technologies into this traditional approach has improved adaptability, efficiency, and safety. Recent advances focus on rock mass classification, where machine learning models interpret drilling parameters for real-time lithology recognition, enabling adaptive excavation strategies (Wang et al., 2021). Blasting quality assessment, including overbreak and underbreak prediction (Hong et al., 2023; Jin et al., 2025), has benefited from ensemble and optimized algorithms, while MWD-based semi-supervised learning addresses data scarcity in blast performance monitoring (Jin et al., 2025; Peng and Wang, 2025), as shown in Fig. 15. Intelligent models have also been applied to hydrogeological risk forecasting, improving prediction of water inflow in heterogeneous conditions (Mahmoodzadeh et al., 2021; Samadi et al., 2025), and to construction cycle time estimation, offering data-driven alternatives to empirical scheduling (Mahmoodzadeh et al., 2025).

Overall, the integration of intelligent technologies in drill-and-blast tunneling is progressively transforming traditional practices into adaptive, data-centric processes. Future directions may include embedding

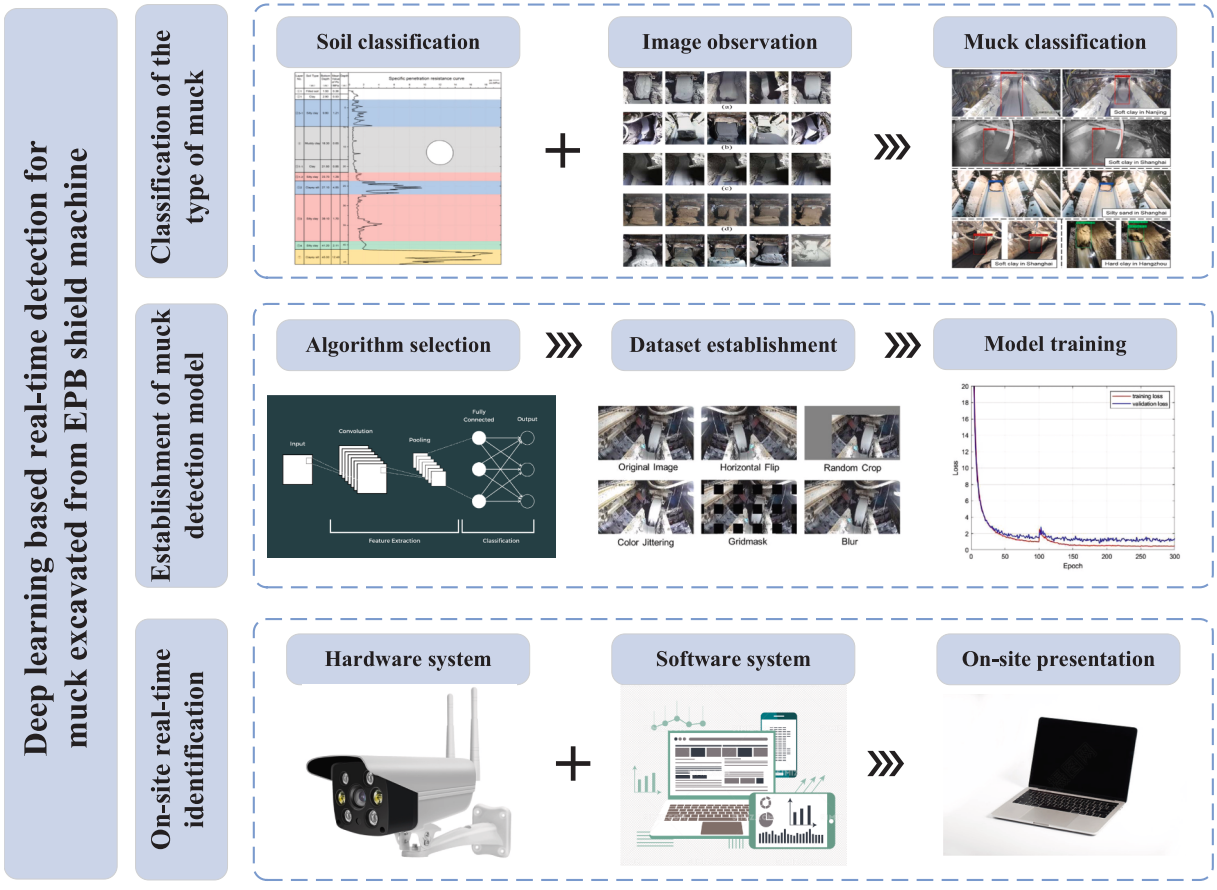

Fig. 14. Framework of CNN-based real-time muck identification (Zhang et al., 2023a).

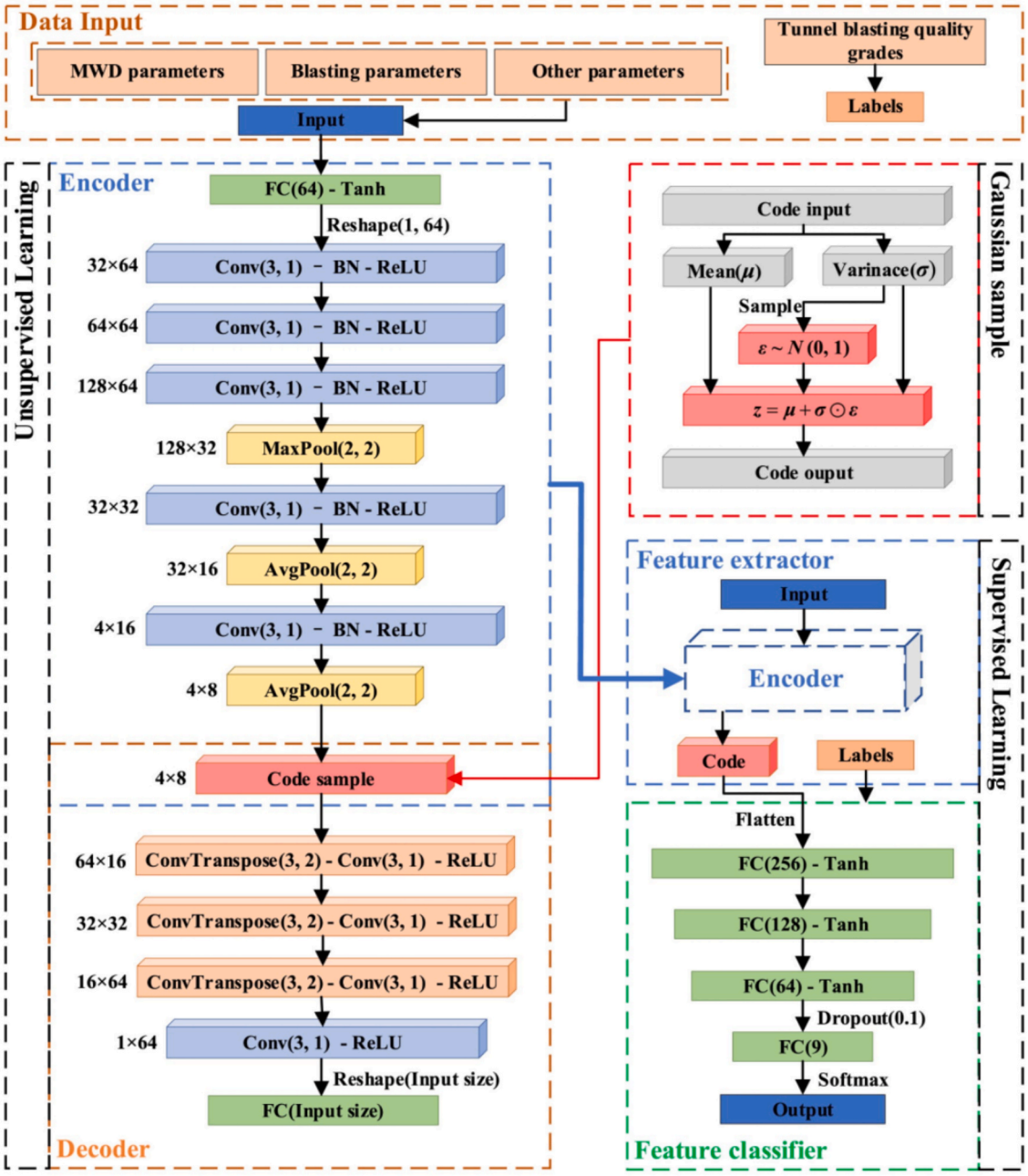

Fig. 15. Flowchart of proposed CVAE + DNN prediction model (Jin et al., 2025).

physics-informed learning, enhancing multi-source sensor fusion, and leveraging real-time feedback loops to further elevate the intelligence level and automation capacity of conventional tunneling methods.

2.3. Application of key technologies in the O&M phase of tunnels

Ensuring the long-term safety and reliability of tunnels during operation and maintenance (O&M) is essential for the overall resilience of underground infrastructure. Potential defects and degradation highlight the need for early fault detection and proactive maintenance strategies. Recent advances in inspection and monitoring technologies, particularly real-time sensing, dynamic condition assessment, and early warning systems play a pivotal role in identifying risks and enabling timely intervention. This section reviews representative studies on tunnel O&M, synthesizing progress in defect detection, continuous monitoring, and intelligent maintenance technologies, based on literature retrieved with keywords such as 'tunnel operation', 'defects detection', 'real-time monitoring', and 'maintenance technologies' across Web of Science.

2.3.1. Defects detection

During the O&M phase, tunnel structures are increasingly exposed to long-term environmental and mechanical effects, making them prone to defects such as cracks, water leakage, segment dislocation, and cross-section convergence. These anomalies not only degrade structural performance but also threaten operational safety, especially in deep-buried tunnels subject to high stress and hydrostatic pressure. Traditional manual inspections often suffer from poor visibility, subjective judgment, and incomplete coverage, highlighting the need for intelligent and automated inspection technologies. Recent advances in this domain can be summarized in three major directions: (i) crack detection, focusing on surface and structural crack recognition; (ii) leakage detection, targeting

water ingress and related hydrogeological risks; and (iii) multi-disease detection and identification, which integrates multiple defect types into unified inspection frameworks. The following subsections review key developments in each category, emphasizing both methodological innovations and engineering applicability.

(1) Crack detection

Cracks in tunnel linings are key indicators of structural deterioration and durability loss, often reflecting stress concentrations, fatigue, or construction defects. Their timely detection is critical for operational safety and effective maintenance. Recent years have witnessed a rapid shift from manual inspection to deep learning–based computer vision, which offers higher accuracy, automation, and scalability. Object detection models built on YOLO variants (YOLOv5, Mini-YOLOv5s, YOLOvX) have shown strong performance in real-time crack localization, with refinements for small-object detection and computational efficiency (Duan et al., 2024; Xue et al., 2024a; Zhou et al., 2023b; Xu et al., 2024a). Complementary segmentation methods, including enhanced DeepLabV3+, Retinex-based networks, and GAN-augmented architectures, improve crack boundary delineation and address challenges of complex lighting and limited labeled data. (Pan et al., 2025; Wu and Zhang, 2023; Zhou et al., 2023c). In addition to surface-level crack detection, Bao et al. (2025) proposed a transfer learning–based YOLOv8 framework for identifying internal voids within reinforced concrete segments. By addressing the challenge of steel bar interference, their method offers an effective solution for concealed defect recognition in shield tunnel linings.

To enhance robustness, ensemble learning and multi-sensor fusion approaches integrate outputs from multiple models or combine imagery with auxiliary spatial information. Examples include mobile imaging platforms coupled with ensemble networks for automated inspections and SLAM-assisted 3D reconstruction of crack distributions, as shown in Fig. 16 (Xu et al., 2024a). System-level implementations such as robotic platforms equipped with onboard detection algorithms have further enabled routine, autonomous tunnel inspections (Xu et al., 2025). System-level implementations of intelligent inspection tools have been developed to support autonomous tunnel maintenance operations. Robotic platforms equipped with temporal crack detection algorithms and onboard crack detection algorithms were deployed for routine inspections (Qin et al., 2023). Pilot applications in metro tunnels have demonstrated the feasibility and scalability of these intelligent inspection solutions under real-world operating conditions (Brien et al., 2023; Lei et al., 2021).

(2) Leakage detection

Water leakage threatens tunnel durability and service safety, particularly in aging or heavily loaded structures. Recent work has moved from manual inspection to automated pipelines that couple visual sensing, 3D reconstruction, and deep learning. Feature-enhanced learning improves the perception of weak, thin, and reflective leakage patterns under adverse lighting (Wang et al., 2024a). Semantic segmentation models, such as Mask R-CNN and improved variants have shown robust performance on lining images and point-cloud data (Liu et al., 2024b; Wang et al., 2024e). To achieve spatially consistent localization, 3D voxel modeling and Structure-from-Motion (SfM) are integrated with CNNs to recover leakage geometry and map defect extent along the tunnel axis (Xue et al., 2022), as shown in Fig. 17.

Beyond single-model detectors, hybrid frameworks that combine classical image morphology (edge extraction, dilation) with CNNs improve boundary fidelity at joints and segment interfaces (Ma et al., 2025). To enhance engineering credibility, visual explanation tools (CAM/Grad-CAM) are embedded to expose attention regions and support result auditing (Feng et al., 2023). Field deployments in metro O&M have verified feasibility for routine inspection; remaining challenges include illumination variability, water-film glare, and domain shift across lines. Priorities include standardized image acquisition, cross-project benchmarks, and tighter coupling with geometric reconstruction to strengthen generalization and maintenance decision-making.

(3) Multi-disease detection and identification

Tunnel structures often exhibit multiple coexisting defects such as cracks, seepage, and lining deterioration, whose combined effects pose greater risks than isolated anomalies. Recent studies employ image-based deep learning to simultaneously detect and segment diverse surface defects in complex subway environments (Huang et al., 2018a; Xu et al., 2023b), as shown in Fig. 18. Vision Transformer (ViT)-based architectures further improve segmentation accuracy by capturing long-range spatial dependencies, while the integration of ground-penetrating radar (GPR) with neural networks enables the identification of hidden subsurface defects beyond the reach of conventional visual inspection (Qin et al., 2024a) (Zhao et al., 2024a).

Multi-source data fusion has emerged as a key strategy to enhance robustness, combining imagery, GPR signals, and construction logs into unified pipelines (Gan et al., 2024), as shown in Fig. 19. Two-stage frameworks that first perform coarse classification followed by fine-grained segmentation further improve localization and characterization accuracy (Feng et al., 2024a). Collectively, these approaches extend tunnel disease assessment from single-defect detection toward comprehensive, multi-modal, and system-level evaluation, supporting more reliable O&M decision-making.

This section reviewed recent advances in defect detection technologies for tunnel O&M, covering cracks, leakage, and multi-disease identification. These methods have greatly enhanced the accuracy and efficiency of tunnel condition assessment, and integrated systems now enable simultaneous detection of multiple defect types. Nevertheless, challenges persist, including label ambiguity, inter-class feature overlap, reliance on large-scale annotated datasets, and limited generalizability across different tunnel types. These issues continue to constrain the robustness and scalability of current models in real-world monitoring applications.

2.3.2. *Tunnel monitoring*

Structural health monitoring (SHM) is critical for ensuring the long-term safety and operational reliability of tunnel infrastructure. Recent progress in sensing technologies, data analytics, and cyber-physical

Fig. 16. Example of tunnel palm line images and detection results using the YOLOv5 algorithm (Xu et al., 2024a).

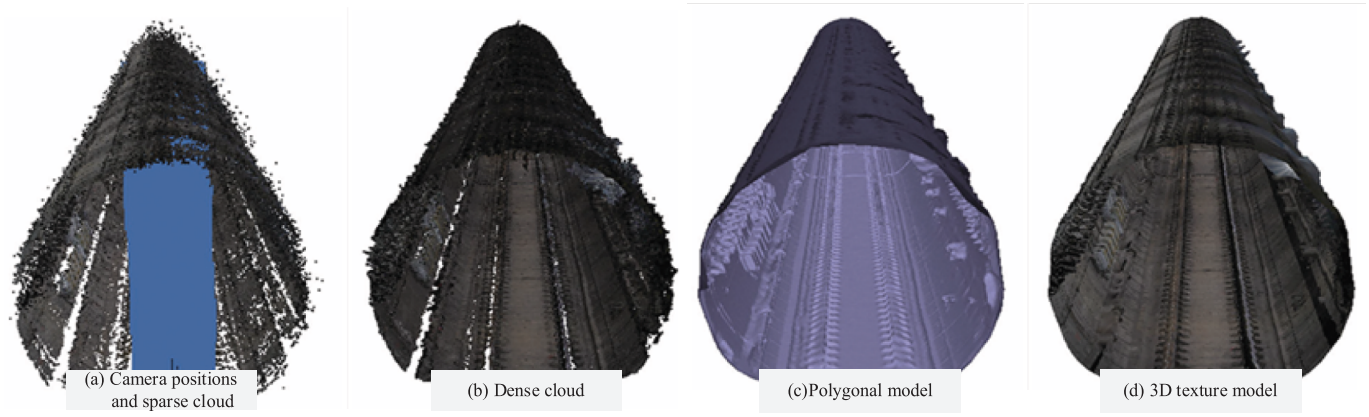

Fig. 17. The result of each link in the 3D tunnel reconstruction process (Xue et al., 2022).

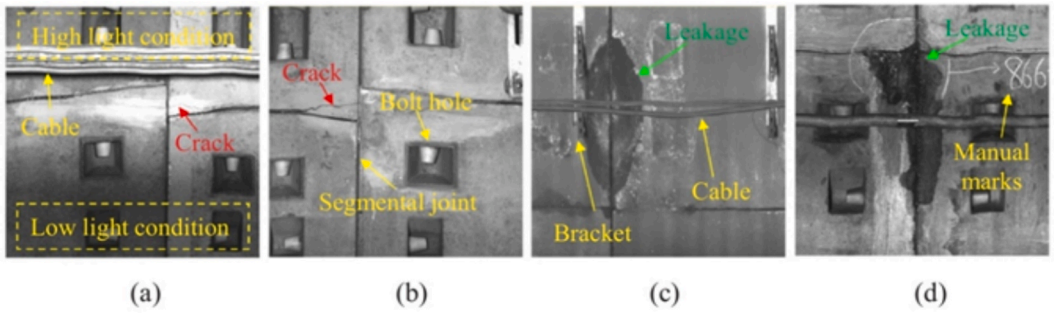

Fig. 18. Acquired lining images of metro shield tunnel (Huang et al., 2018a).

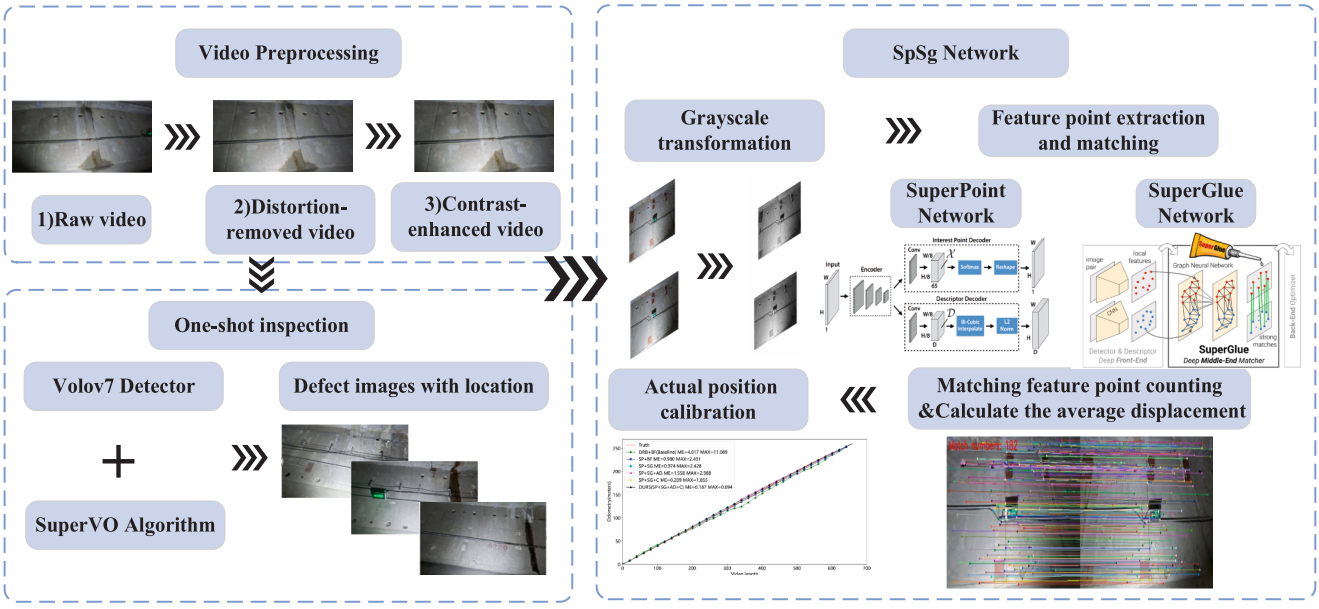

Fig. 19. Image matching process based on the SpSg Network (Gan et al., 2024).

systems has substantially improved the precision, automation, and interpretability of SHM practices in tunneling. Machine learning and deep learning approaches have been widely adopted to predict structural responses based on real-time SHM data. For example, LSTM models have been utilized to forecast deformation and estimate additional stress induced by adjacent tunneling operations (Ye et al., 2024). Moreover, ensemble learning frameworks have demonstrated enhanced predictive accuracy for mechanical behavior in complex environments, such as underwater shield tunnels, through long-term SHM datasets (Jin et al., 2023; Tan et al., 2023), as shown in Figs. 20-21.

To further support predictive maintenance and resilience planning, Wu et al. (Wu et al., 2025) developed a GNN-based framework that models the spatial propagation of seismic damage in shield tunnels with contact loss, offering improved sensitivity to defect evolution. Shen et al. (Shen et al., 2025a) introduced a cloud model-based resilience assessment approach that integrates multiple damage indices derived from

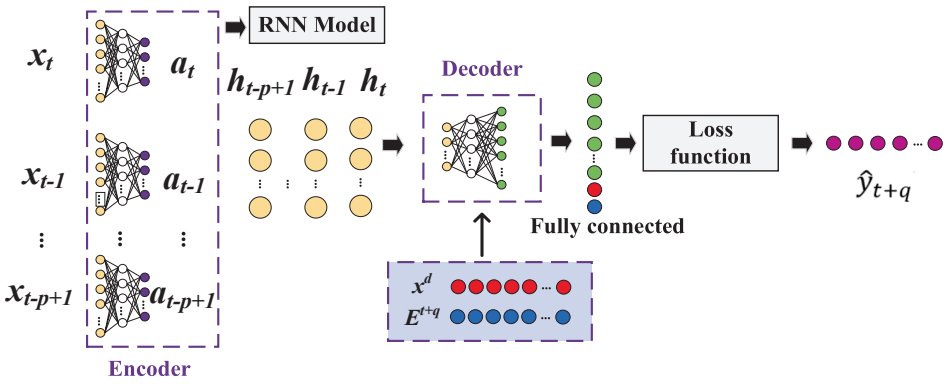

Fig. 20. Framework of multivariate prediction model: ATENet (Tan et al., 2023).

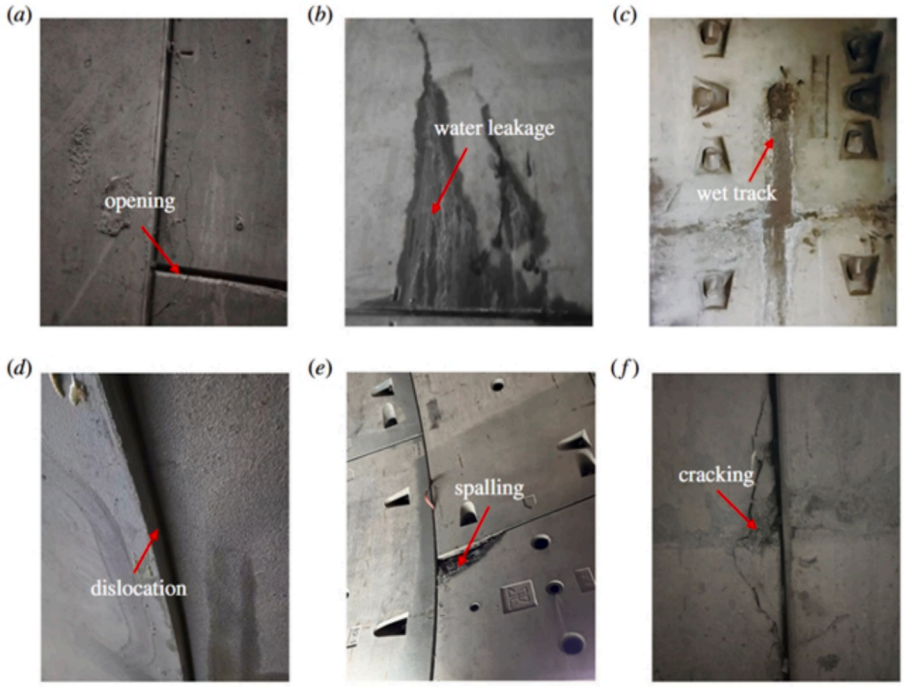

Fig. 21. Diseases caused by opening and dislocation of the segments: (a) opening; (b) water leakage; (c) wet track;(d) dislocation; (e) spalling and (f) cracking (Jin et al., 2023).

SHM data, enabling probabilistic interpretation of tunnel vulnerability under seismic events. Additionally, Shen et al. (Shen et al., 2025b) proposed a resilience modeling strategy that incorporates bolt preload loss and multistage functionality degradation curves, providing insight into mechanical performance loss during the tunnel lifecycle.

Distributed optical fiber sensors have been widely adopted for real-time monitoring of stress and deformation in both underwater and submarine shield tunnel, offering high-resolution, long-distance, and continuous sensing capabilities (Bao et al., 2024; Mao et al., 2011; Yang et al., 2023a). Non-contact methods, such as mobile laser scanning and binocular vision systems, have demonstrated high spatial accuracy for tunnel cross-sectional deformation tracking under dynamic operational conditions (Camara et al., 2024; Yue et al., 2024). When integrated with mobile robotic platforms, these vision-based approaches enable autonomous acquisition and spatiotemporal mapping of tunnel geometry changes (Chai et al., 2024; Jeong et al., 2023).

However, several challenges remain, including limited robustness across tunnel types, fragmented data interoperability, and the difficulty of scaling AI models under diverse field conditions. Emerging approaches such as blockchain have been proposed for secure and transparent SHM data management (Zhang et al., 2024a), though their practical value still requires validation. Future research should prioritize cross-modal sensing fusion, explainable AI, and lifecycle-integrated SHM frameworks to enhance adaptability, transparency, and engineering applicability of intelligent monitoring systems.

3. Platform architecture evolution

With the continuous advancement of information technology, intelligent sensing, and analytical modeling methods, platform architecture in the field of tunnel engineering has undergone a profound transformation. This evolution can be divided into three stages, as shown in Fig. 22:

Single-function architecture (pre-2015): Initially, tunnel platforms were designed for isolated tasks such as data acquisition or equipment monitoring, offering limited support for interdisciplinary coordination and decision-making.

Integrated-function architecture (2015–2020): With the growing demand for life-cycle management and operational efficiency, platforms began to integrate core technologies such as BIM, IoT, and GIS, enabling

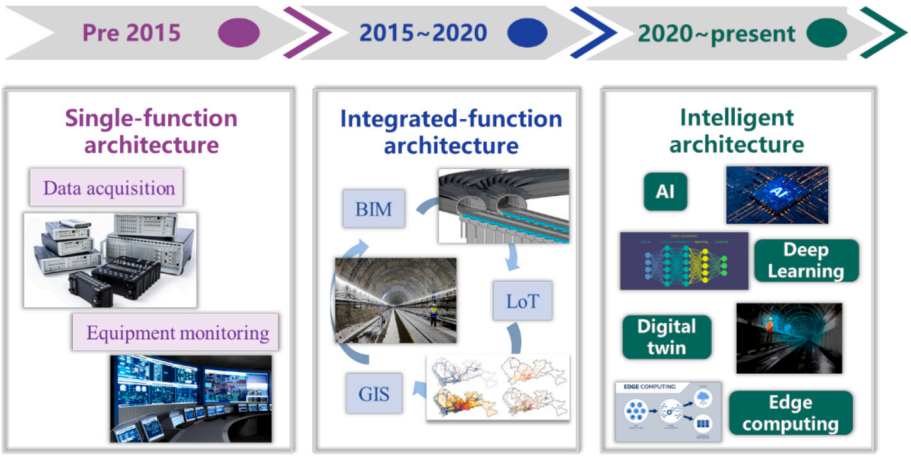

Fig. 22. Evolution of Tunnel Platform Architectures Across Time and Functionality.

multi-source data fusion and facilitating collaboration across design, construction, and O&M phases (Liu et al., 2023c; Liu et al., 2025b).

Intelligent architecture (2020-present): In recent years, the emergence of AI, digital twin models, deep learning, and edge computing has significantly accelerated the development of intelligent platforms. These platforms are now equipped with real-time sensing, predictive analytics, and autonomous control capabilities (Li et al., 2024). For instance, digital twin models have been developed to dynamically evaluate tunnel structural performance throughout the service lifecycle (Zhou et al., 2024a), while service-oriented architecture (SOA)-based control platforms support construction safety management through the fusion of multi-source monitoring data (Mao et al., 2024). Additionally, intelligent robotic systems designed for deep-buried and long-distance tunnels are being integrated with environmental perception and autonomous decision-making modules, enabling platform support for automated operations in complex underground scenarios (Zhang et al., 2023c).

This section reviews the evolutionary process of platform architecture, examines the functional characteristics across different development stages, and discusses the enabling roles of these platforms in tunnel operation and maintenance (O&M). Furthermore, it explores future directions and potential applications of intelligent platform systems in advancing tunnel lifecycle management.

3.1. *Single function architecture*

The earliest digital platforms in tunnel engineering were characterized by a single-function architecture, designed primarily to support isolated tasks within specific phases of a project, such as data acquisition, equipment status monitoring, or progress visualization. These systems typically operated in silos and lacked interoperability and data integration, resulting in fragmented workflows and inefficiencies across the tunnel lifecycle. This stage aligns with the initial phase of BIM evolution, described by Brkić as the 'lonely BIM' era, where digital tools served individual disciplines without cross-domain collaboration or integrated workflows (Borkowski, 2023). Essentially, these platforms functioned as digitized replicas of manual processes, limited to static modeling or documentation-oriented tasks.

In tunnel engineering, typical examples include standalone platforms for monitoring ground settlement, logging TBM data, or documenting inspections. For instance, Bossi (Bossi et al., 2017) developed a specialized monitoring system for structural health assessment of a road tunnel intersecting an active landslide, focusing exclusively on localized deformation and strain data. Similarly, Qin (Qin, 2019) proposed a dedicated monitoring platform for submarine cable tunnels, which applied data mining techniques to support operational decision-making. However, these platforms remained confined to single-domain applications and lacked integration with broader tunnel life-cycle management frameworks. While effective within their specific scopes, they were not designed for interoperability, data sharing, or system-level coordination, thereby limiting their value in holistic tunnel management. Despite their limitations, single-function platforms served as a critical role for subsequent digital development. They facilitated the initial introduction of digital tools into engineering workflows, highlighted the inefficiencies of fragmented and non-interoperable systems, and ultimately catalyzed the shift toward integrated-function and intelligent platform architectures.

3.2. *Integrated function architecture*

Driven by the growing demand for efficient and holistic tunnel management, the architecture of tunnel platforms has progressively evolved from single-function systems to integrated-function frameworks. This transformation has been facilitated by the convergence of enabling technologies, such as cloud computing, big data analytics, and IoT, which collectively support cross-domain integration and intelligent decision-making. While these technologies address different functional needs, their integration substantially enhances the overall performance of tunnel management platforms by breaking down data silos and enabling interoperability across construction and O&M stages. Through the adoption of unified data standards and advanced integration interfaces, platforms can now seamlessly link shield tunnelling parameters, geological data, monitoring information, and structural health data, creating a unified and comprehensive data ecosystem that supports collaborative operations and informed decision-making.

(1) Multi-scale Tunnel Modeling and Collaboration Platform, as shown in Fig. 23 (Borrmann et al., 2014). As a core component of the integrated function architecture, the platform supports seamless operations across multiple project scales from master planning to detailed designing. By employing process-oriented modeling techniques, it ensures consistency across various levels and supports automatic updates, thereby enhancing both design accuracy and operational efficiency. Moreover, the platform architecture facilitates temporal collaboration among multidisciplinary teams, enabling simultaneous editing and coordination within shared model environments. Its high compatibility with heterogeneous design tools further promotes cross-domain integration, supporting a unified and coherent framework for tunnel project delivery.
(2) SATBIM platform, as shown in Fig. 24 (Ninic et al., 2017). This platform integrates tunnel information modeling (TIM), structural analysis, and visualization functions into a unified

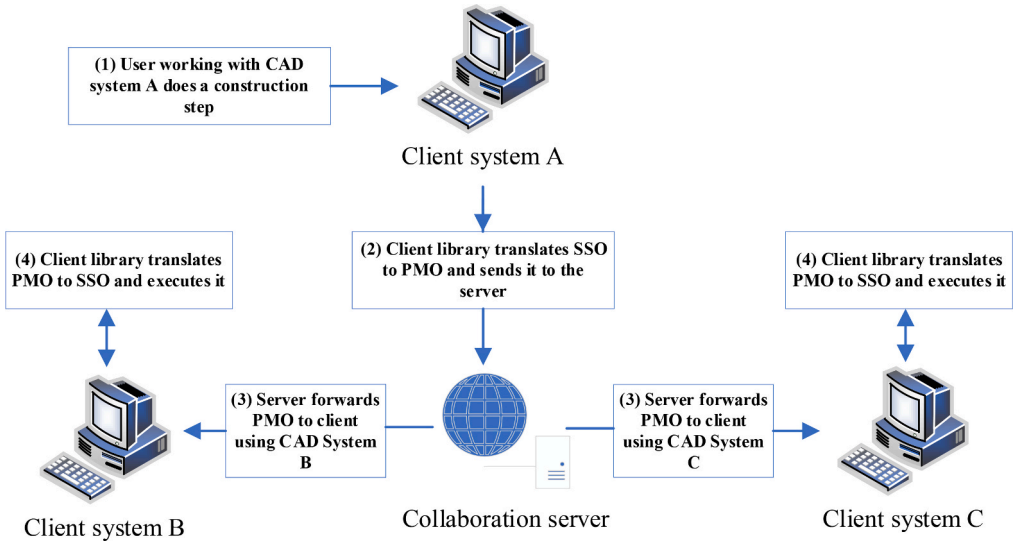

Fig. 23. Synchronous collaborative modeling based on a system-independent procedural framework (Borrmann et al., 2014).

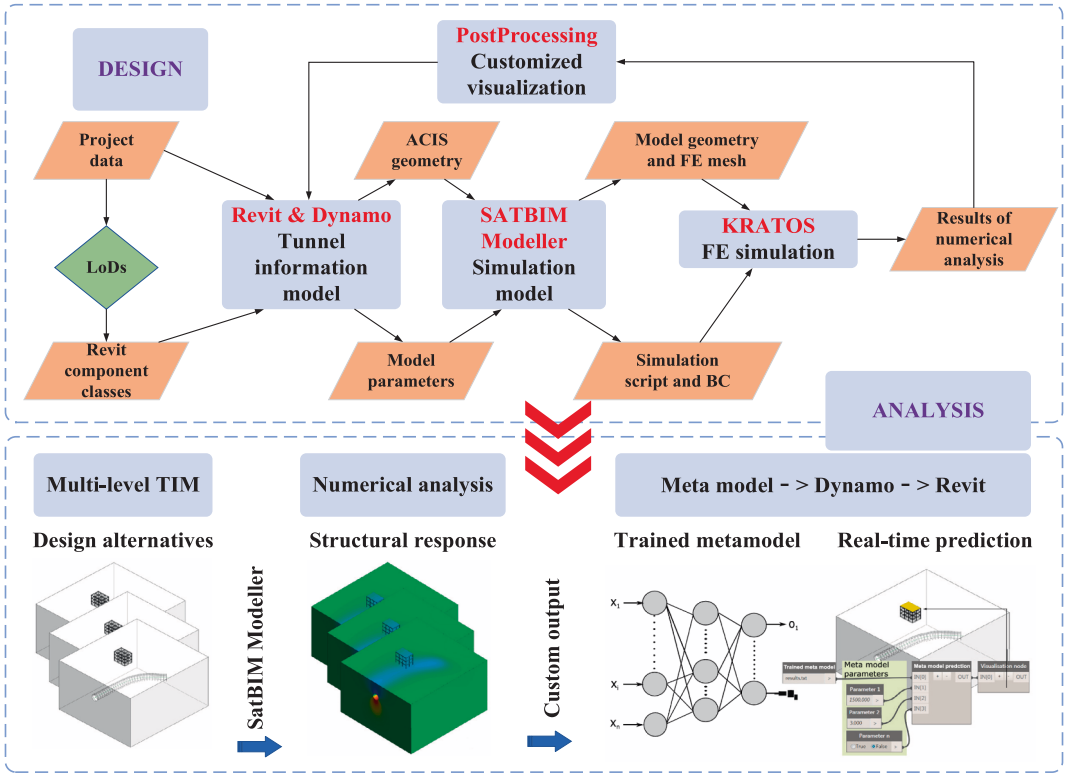

Fig. 24. Flowchart of the SATBIM platform (Ninic et al., 2017).

workflow. It automatically generates simulation-ready tunnel models through multi-level tunnel information components, which are subsequently analyzed using the open-source simulation engine KRATOS. SATBIM adopts a modular architecture, enabling both the generation of structural models during the design phase and the performance of soil-structure interaction analyses during construction.

(3) Construction data collection system, as shown in Fig. 25 (Huang et al., 2018b). As a key component supported by the integrated function architecture, this platform enables automated data modeling and real-time visual analytics to evaluate working face stability during shield tunneling. By integrating monitoring data such as support pressure and ground settlement, the system assists engineers in performing comprehensive assessments and optimizing shield tunneling parameters in real time. This integrated approach enhances construction and significantly improves risk management throughout the excavation process.

(4) Tunnel boring machine test platform, as shown in Fig. 26 (Xie et al., 2025b). The platform comprises several subsystems, including the tunneling execution subsystem, hydraulic loading subsystem, specimen mounting subsystem, control subsystem, and integrated monitoring subsystem. These modules work in coordination to simulate a wide range of geological conditions and tunneling scenarios, enabling in-depth investigation of rock-

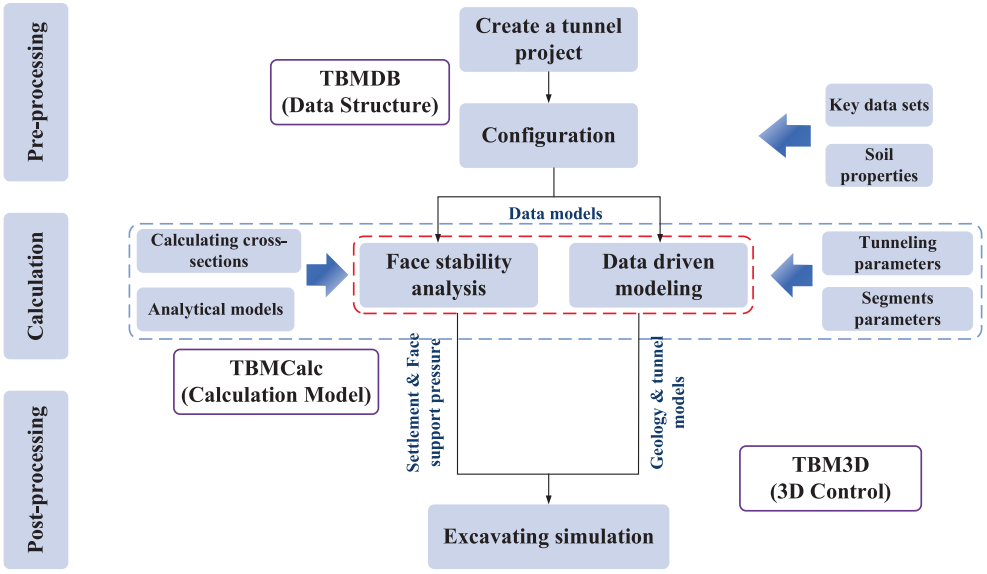

Fig. 25. Flowchart of tunnel face stability analysis in TBM Studio (Huang et al., 2018b).

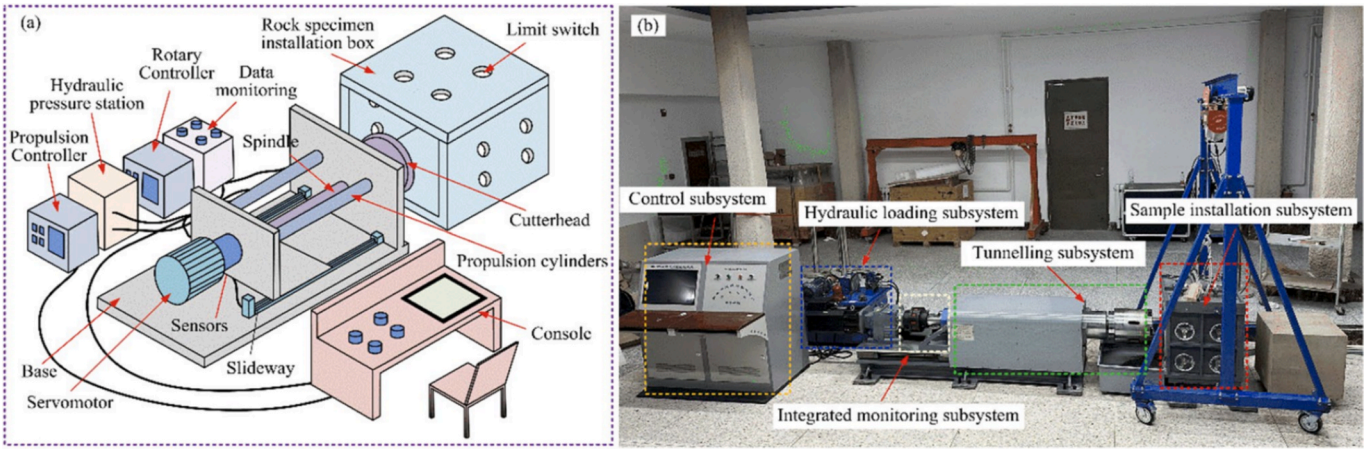

Fig. 26. The novel TBM tunnelling test platform for rock-machine interaction analysis. a Schematic diagram of the platform. b Actual setup housed at Tsinghua University, China (Xie et al., 2025b).

machine interactions. The platform supports performance evaluation and parameter optimization of shield machines, thereby contributing to safer and more efficient tunneling operations.

(5) UT-SIM platform, as shown in Fig. 27 (Sayed et al., 2019). This platform integrates two key subdomains: the soil domain, simulated using OpenSees through open-source finite element analysis framework, and the tunnel domain, which modeled with VecTor2 as a nonlinear finite element tool specialized for reinforced concrete analysis. The UT-SIM framework has been applied to simulate the structural response of a subway tunnel during the 1995 Great Hanshin (Kobe) Earthquake, enabling in-depth evaluation of soil-structure interactions and nonlinear failure mechanisms under seismic loading.

3.3. Intelligent architecture

Tunnel digital intelligent construction and O&M rely on the integration of information technology, computational systems, sensor networks, and AI, augmented by big data analytics, to enable intelligent management across the entire tunnel lifecycle. This intelligent management organically integrates multi-source information across disciplines into a cohesive digital platform, breaking down traditional functional limitations and enabling temporal information sharing, feedback coordination, and collaborative decision-making throughout all phases.

(1) Multi-sensor information sensing and integrated management platform, as shown in Fig. 28 (Zhang et al., 2023c). This platform integrates communication networks, cloud computing, and advanced geotechnical sensing technologies to enable comprehensive multi-source data acquisition and real-time decision optimization throughout the entire tunnel boring process. By embedding a particle swarm optimization (PSO) algorithm, the system dynamically identifies the optimal construction scheme that minimizes tunnelling specific energy (TSE), thereby significantly improving construction efficiency, enhancing safety, and supporting adaptive control under complex geological conditions.

(2) Tunnel modeling and optimization design platform, as shown in Fig. 29 (Ye et al., 2023). This platform is developed based on C++ programming and tightly integrated with the 3DEC discrete element analysis environment. By consolidating core functions into an intuitive interface, the platform significantly lowers the operational complexity of discrete element method (DEM)

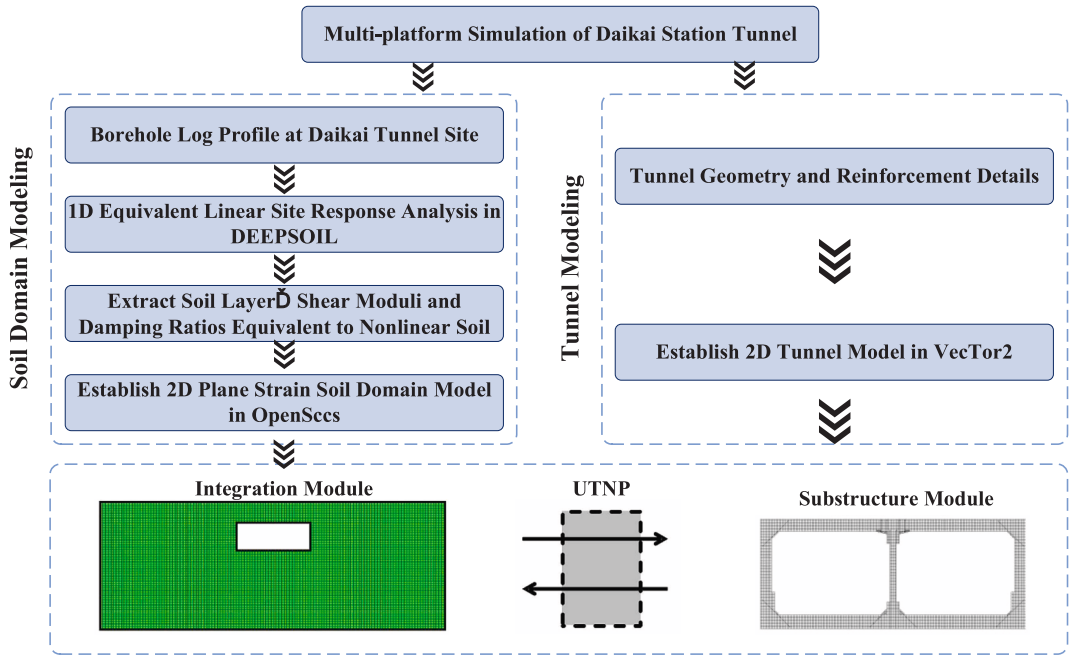

Fig. 27. Multi-platform simulation procedure flowchart (Sayed et al., 2019).

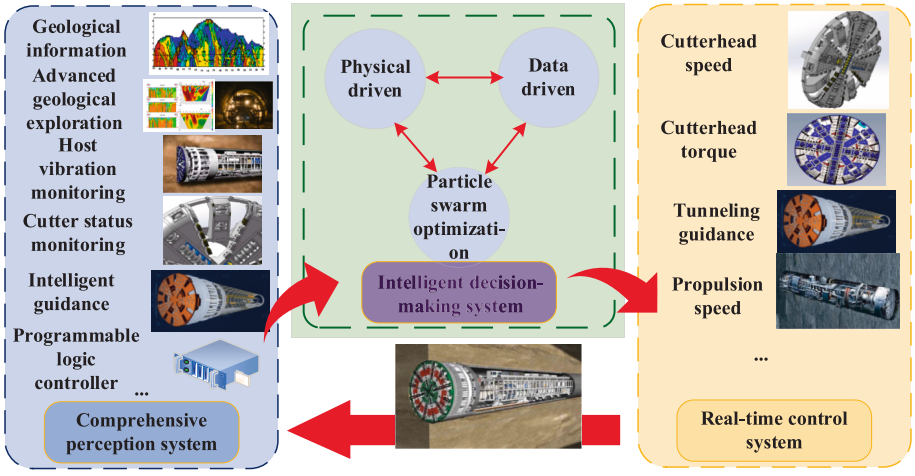

Fig. 28. Overall structure diagram of the intelligent tunnelling robot system (Zhang et al., 2023c).

simulations, enhancing usability for geotechnical and structural engineers engaged in tunnel optimization design.

(3) Multi-information intelligent warning and safety management platform, as shown in Figs. 30-31 (Zhou et al., 2024a). Based on digital twin technology, the platform is designed to enhance temporal safety risk control during tunnel construction. By deploying a multifunctional communication base station, it enables the automated acquisition and transmission of heterogeneous data, including construction parameters and surrounding rock response. The platform performs virtual simulations based on dynamic datasets and integrates a four-stage warning system along with a predefined emergency response protocol, supporting real-time condition assessment and proactive decision-making. In the case of the deeply buried Dongtianshan Tunnel, the platform successfully predicted and alerted a potential excavation face collapse, thereby ensuring timely mitigation and safeguarding personnel safety.

While the platform has demonstrated practical effectiveness in emergency scenarios such as the Dongtianshan Tunnel face collapse warning, current implementations lack standardized latency performance metrics. In this case study, the estimated time from abnormal data acquisition to alarm issuance was within 15 s, primarily constrained by the data pre-processing and model inference cycles at the edge computing unit. However, comprehensive benchmarks, such as end-to-end latency under various load conditions, emergency response actuation times, and communication jitter rates are still lacking across most systems. Future development should incorporate latency-aware design principles and simulation-based stress testing to ensure reliability under extreme operational demands.

(4) Smart metro service platform, as shown in Fig. 32 (Pan and Zhang, 2022). This platform facilitates continuous and dynamic monitoring of metro tunnel structural health by integrating data acquisition, early warning mechanisms, and maintenance management modules. By collecting multi-sensor data in real time, the system performs operational condition analysis, promptly identifying anomalies and early warning signals. Equipped with

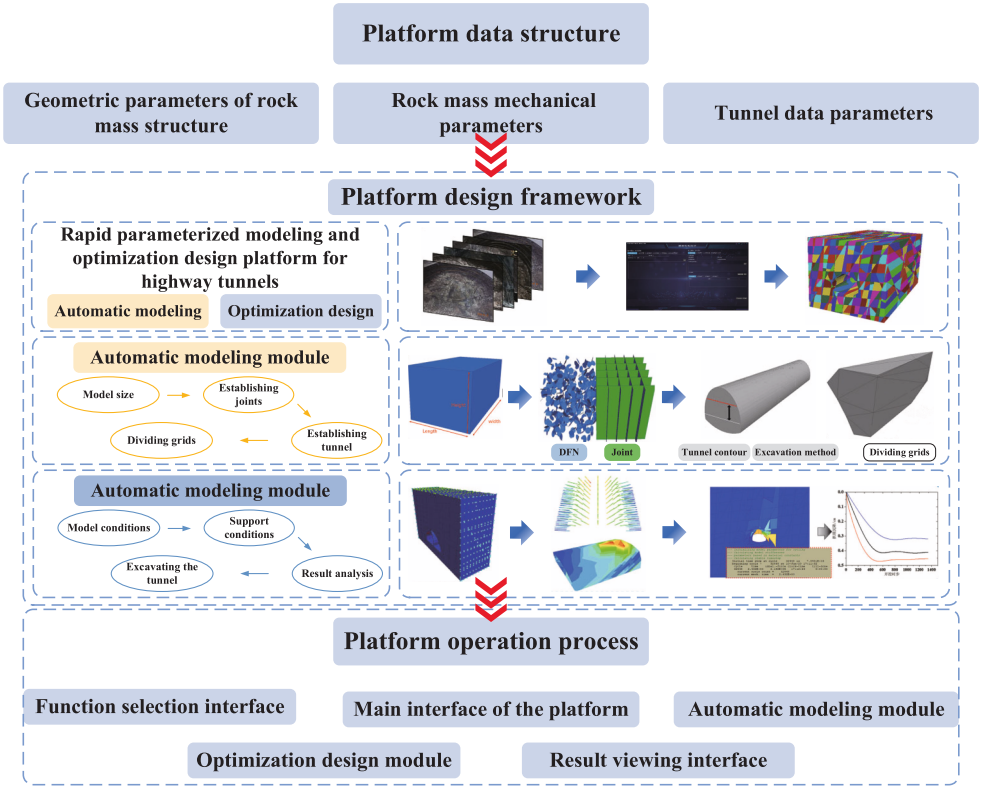

Fig. 29. Framework of the platform (Ye et al., 2023).

an intelligent decision support system, the platform enables the O&M strategies, improves resource allocation efficiency, and reduces failure rates, thereby significantly enhancing the reliability and safety of metro tunnel operations.

(5) DNN based risk prediction and optimization platform, as shown in Fig. 33 (Zhang et al., 2021b). This platform establishes a risk prediction and control optimization framework by integrating deep neural networks (DNN) with genetic decision optimization (GDO) algorithms. The DNN model is employed to capture complex nonlinear relationships between key influencing factors and risk outcomes in tunnel construction. By combining multi-objective techniques, the platform generates optimal risk mitigation strategies, enabling accurate early warnings and real-time adaptive adjustments. This integrated approach significantly improves both construction safety and operational efficiency under dynamic and uncertain conditions.

4. Challenges and prospects

4.1. Challenges

Intelligent tunnel construction technologies and O&M platforms have achieved significant advancements, particularly in multi-source data acquisition, real-time monitoring, risk forecasting, and intelligent decision support. However, with the rapid expansion of tunnel engineering, especially the development of deep and complex underground spaces in urban areas, the growing need for life-cycle maintenance of aging infrastructure, and the increased emphasis on low-carbon and sustainable construction, the limitations of conventional technologies and management approaches have become increasingly evident. These traditional often fall short in addressing the challenges posed by high-risk geological conditions, intricate engineering scenarios, and evolving safety and efficiency requirements. This gap highlights the urgent need for integrated, adaptive, and intelligent solutions capable of supporting resilient tunnel construction and operation under dynamic and uncertain conditions.

(1) The lack of effective data integration and interoperability remains a major barrier to intelligent tunnel management. The tunnel construction and O&M process generates large volumes of heterogeneous data, including geological surveys, construction monitoring, and equipment operation records, which often exist in incompatible formats and lack standardized interfaces. As a result, integrating and sharing these data across systems becomes difficult, reducing the effectiveness of intelligent tools, particularly for real-time decision-making and cross-phase data continuity. A case study (Feng et al., 2024b) from the Changsha Metro Line 4 demonstrated how an AIoT-based system effectively integrated machine parameters and settlement data to support predictive control. However, the system's performance was highly dependent on localized datasets, and the machine learning models required retraining before use in other tunnel projects, exposing the broader limitations in data standardization and transferability.

(2) The current level of intelligence embedded in tunnel construction and O&M platforms is still insufficient to meet real-world demands. Despite the integration of automation, AI, and IoT technologies, there remains a clear gap between research progress and field application. AI-based prediction models are constrained by limited training data and often struggle to maintain accuracy in highly variable geological conditions. For example, deformation prediction models (Sauer and Fischer, 2013) trained on small-scale datasets require frequent retraining to remain effective in different tunneling scenarios. Similarly, although digital twin systems hold promise, they are hampered by slow model updates and latency, which limit their utility for real-time control. This becomes especially critical under extreme or emergency conditions, where delayed system responses may compromise

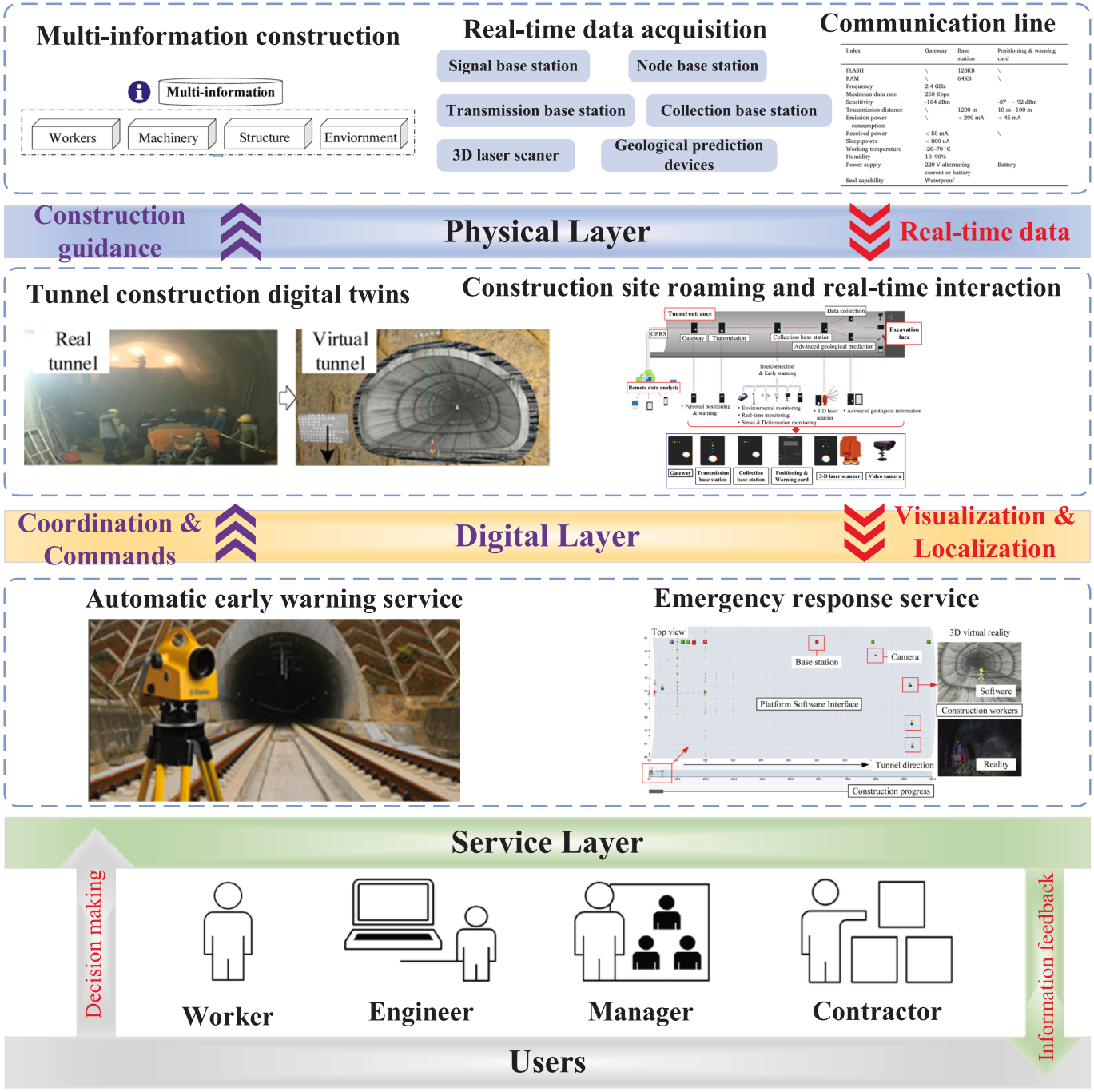

Fig. 30. DT-based platform framework (Zhou et al., 2024a).

safety or reduce the effectiveness of early warnings. Moreover, current platforms lack standardized latency evaluation metrics, and their real-time monitoring and actuation capabilities remain under-validated in field-scale applications. These issues collectively underscore the need for robust algorithms, scalable datasets, empirical latency benchmarking, and enhanced computing capacity to realize adaptive, intelligent tunnel systems that can operate reliably under dynamic, high-risk scenarios.

(3) The lack of sufficient system security and operational robustness poses serious risks to intelligent platform deployment in hazardous environments. As tunnel construction increasingly occurs in high-risk settings such as areas prone to earthquakes, flooding, or extreme weather, platforms face growing threats from cyber-attacks, data breaches, and system disruptions. Ensuring platform security and maintaining data reliability are critical prerequisites for the broader adoption of intelligent construction systems. Without strong cybersecurity measures and system resilience, platform trustworthiness is undermined, thereby limiting their ability to support safety–critical decision-making and reliable tunnel operation.

(4) The inherent limitations of current algorithms and the poor generalization capability of AI models hinder the practical scalability of intelligent tunnel systems. Although machine learning and deep learning have made notable strides in tunnel engineering applications, most models still perform well only under specific tasks or datasets. They often fail to generalize across varying geological environments, construction methods, or monitoring systems. In addition, common issues such as data

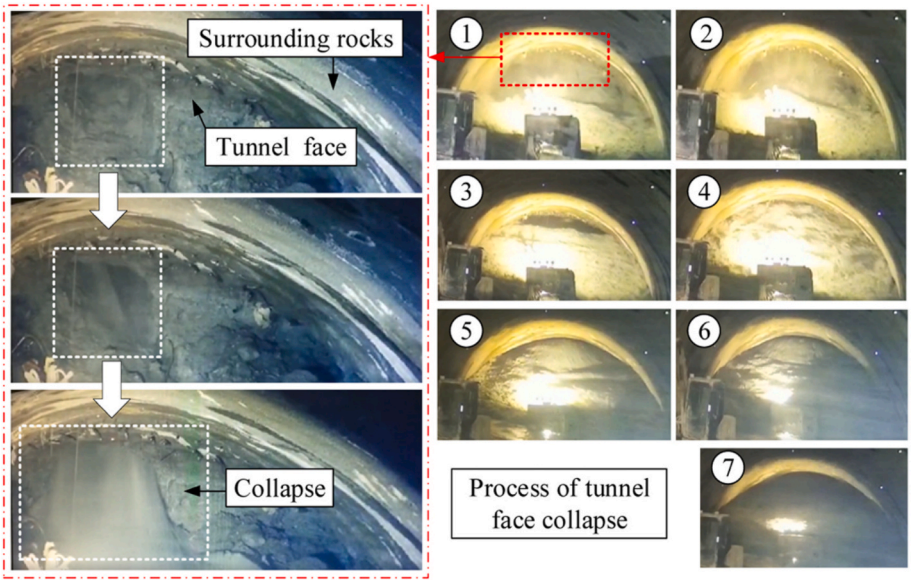

Fig. 31. Collapse of tunnel face recorded in the remote video camera of the system (Zhou et al., 2024a).

imbalance, noise interference, and a lack of labeled data impede model robustness. The low interpretability of complex deep models further restricts their acceptance in safety-sensitive engineering decisions. These challenges collectively constrain the real-world deployment and scalability of intelligent tunnel platforms.

(5) In addition to technical challenges, sustainable and low-carbon construction practices in tunneling require further quantification and engineering-scale validation. Recent efforts focus on resource reutilization and green soil conditioning. For instance, Wu et al. (Wu et al., 2024a) reported in-situ trials on the Suzhou Metro Line 11 project, where EPB shield muck was reused to partially replace synchronous grouting materials. The modified grout achieved comparable mechanical strength while reducing virgin material demand, with spoil reuse rates exceeding 30 %. In another case, Wang et al. (Wang et al., 2024f) investigated bentonite-silty clay modified slurry using field samples from the Shenyang Metro Line 6, deriving quantitative rheological thresholds (yield stress ≈ 120 Pa, slump ≈ 18 cm) under different injection ratios. These studies provide measurable indicators such as spoil reuse percentage, rheological performance ranges, and optimized injection ratios, offering engineering guidelines for sustainable tunneling practices. Nevertheless, wider implementation is constrained by the absence of standardized carbon accounting frameworks and limited demonstration projects. Future work should therefore establish lifecycle-based carbon footprint models and develop benchmark systems supported by policy incentives to promote large-scale adoption.

4.2. Prospects

In future, the intelligent platforms will evolve into integrated, closed-loop ecosystems that directly support tunnel engineering management strategies. By combining real-time sensing, adaptive control, and predictive analytics with digital twin co-simulation, these platforms will not only enhance technical performance but also enable data-driven O&M strategies, resource optimization, and emergency coordination. Furthermore, the embedding of low-carbon indicators, such as carbon footprint tracking, material recycling, and lifecycle sustainability assessment will allow managers to incorporate environmental objectives into routine decision-making. From an industrial perspective, the establishment of standardized data protocols, interoperable modular platforms, and open-source toolchains will lower the barriers to adoption and promote cross-project management integration. Collectively, these developments illustrate how intelligent platforms will facilitate smarter, greener, and more effective engineering management in tunnel construction and O&M.

5. Conclusion

This review systematically examined the evolution of intelligent technologies and platforms across the full life cycle of shield tunnel engineering, encompassing design, construction, and operation & maintenance stages. The study not only highlights the progress in automation and digitalization but also identifies current limitations and outlines future research opportunities to promote resilient and sustainable tunnel systems. Key conclusions are as follows:

(1) Intelligent technologies have been increasingly integrated across the full lifecycle of shield tunneling. They are now widely applied in the design, construction, and operation phases, forming a closed-loop system that encompasses parametric modeling, automated construction, intelligent sensing, and preemptive response. These technologies demonstrate strong coordination between stages and provide a solid foundation for achieving digital and refined lifecycle management of tunnel infrastructure.

(2) Tunnel management platforms are evolving from simple data integration functions, such as visualization and archiving, to intelligent systems capable of real-time analysis, decision-making, and coordinated control. Supported by cloud-edge collaboration, BIM, and IoT technologies, these platforms enable dynamic data fusion and provide unified management across all phases of tunnel projects.

(3) Lifecycle-intelligent tunnel systems still face a series of key engineering challenges related to data integration, algorithm generalization, real-time responsiveness, platform security, and sustainable implementation. Fragmented and non-standardized data constrain cross-phase interoperability, limiting the effectiveness of digital tools in supporting continuous lifecycle management. Although AI models have shown strong potential, most remain project-specific with limited transferability across varying geological strata, and they often suffer from sensor noise, data heterogeneity, and high computational demands. These factors undermine the robustness and scalability of intelligent systems in

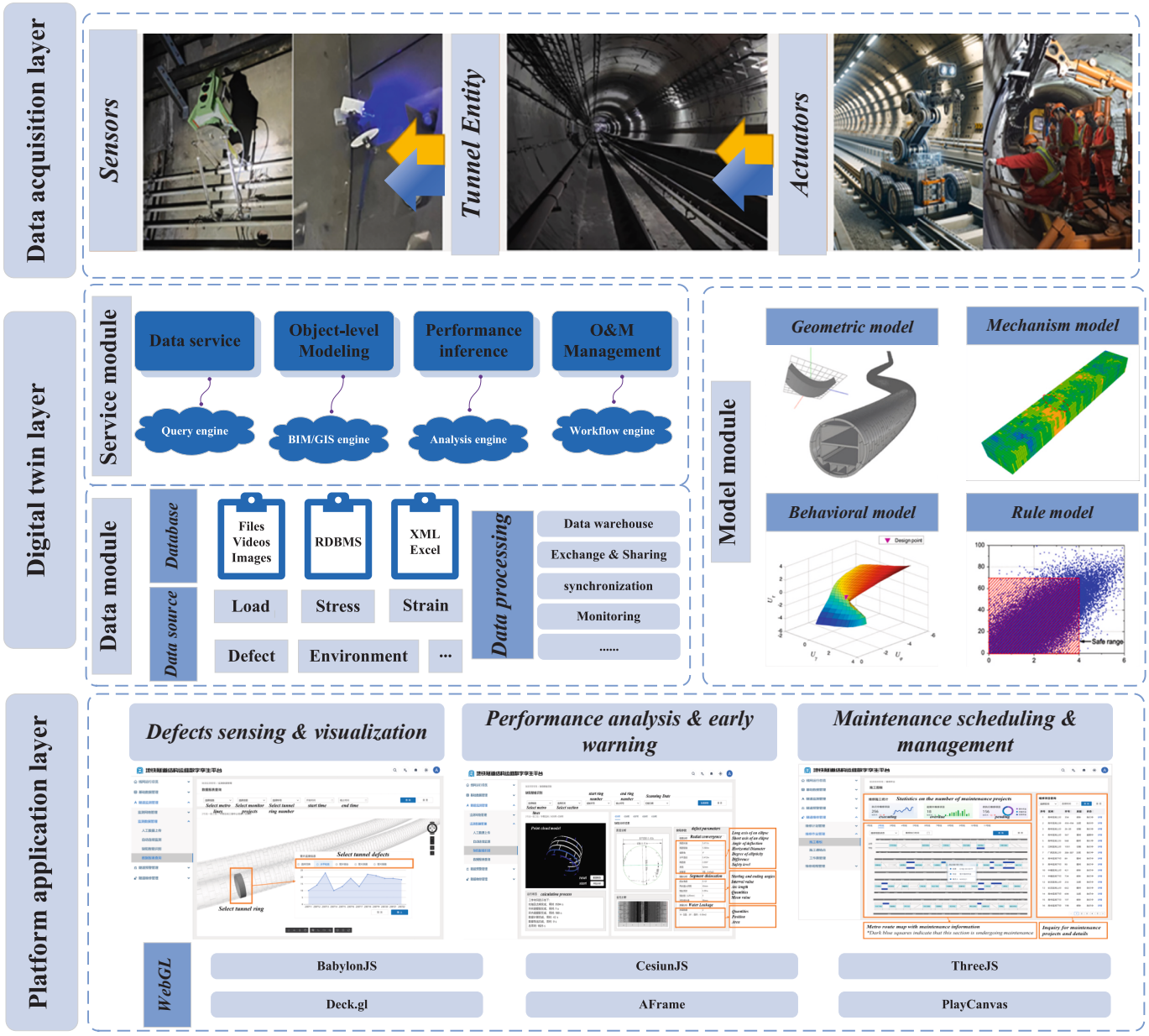

Fig. 32. Smart metro service platform framework (Pan and Zhang, 2022).

real-world conditions. Moreover, the "black-box" nature of deep learning models restricts their acceptance in field operations, especially in safety–critical environments. Real-time control capabilities are further limited by latency and insufficient empirical validation at engineering scale, particularly under high-risk or extreme scenarios. Platform-level security vulnerabilities also pose threats to system stability and data integrity. In parallel, efforts toward sustainable and low-carbon tunneling are hindered by the absence of lifecycle carbon footprint standards and the lack of large-scale demonstration projects. Overcoming these interlinked bottlenecks is essential for advancing toward resilient, adaptive, and environmentally responsible intelligent tunnel systems.

(4) In future, the development of lifecycle-intelligent tunnel systems should advance toward an integrated, resilience-oriented, and sustainable framework. Priority must be given to cross-phase data interoperability and risk-informed decision-making, supported by unified platforms that combine intelligent sensing, automated construction, and real-time operation control. Embedding disaster resilience modeling, carbon footprint evaluation, and intelligent emergency management will extend tunnel platforms from engineering tools to strategic components of urban resilience. Ultimately, intelligent tunnels are expected to converge with broader urban infrastructures, evolving into adaptive, low-carbon, and synergistic subsystems that underpin smart and sustainable cities.

CRediT authorship contribution statement

Xiaohua Bao: Writing – review & editing, Supervision, Methodology, Conceptualization. **Bowen Zheng:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jun Shen:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Xiangsheng Chen:** Supervision, Project administration, Conceptualization. **Heqi Kong:** Writing – original draft, Methodology, Conceptualization. **Junhong Li:** Writing – original draft, Methodology, Conceptualization. **Xiaoqing Wang:** Project administration, Methodology, Investigation. **Hongzhi Cui:**

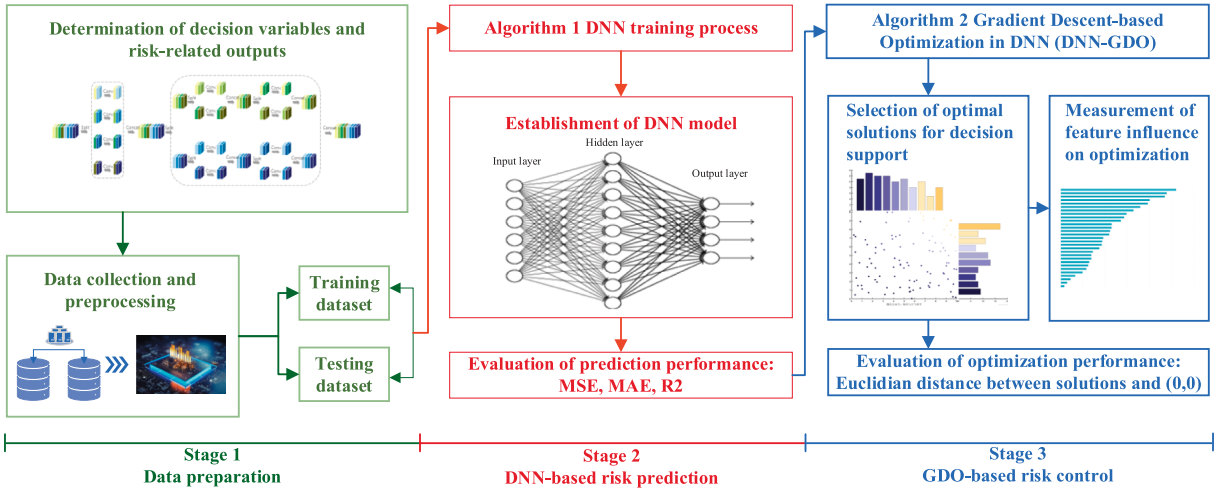

Fig. 33. Proposed DNN-GDO-based framework for decision support in controlling risk in tunnel excavation (Zhang et al., 2021b).

Visualization, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The work described in this paper was supported by National Key Research and Development Program of China (2024YFF0507902), National Natural Science Foundation of China (No. 52525805 & 52578479).

Data availability

No data was used for the research described in the article.

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